

Solid Phase Synthesis Using Organometallic Reagents

H. Gradén and N. Kann*

Department of Chemical and Biological Engineering, Chalmers University of Technology, SE-41296 Göteborg, Sweden

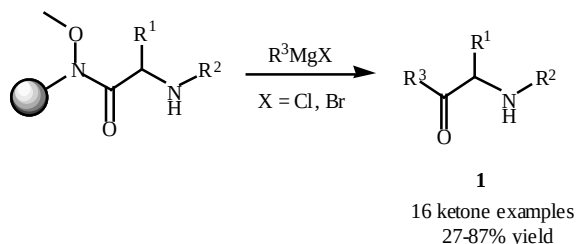
Abstract: The scope of organometallic reactions on solid phase has lately expanded to include many new reactions. In this review, we summarise the recent developments in this area, covering reactions such as metathesis, carbene insertion, radical-mediated transformations, the use of organomagnesium, organozinc and organocopper reagents, applications of titanium alkylidenes for alkene synthesis, as well as the Pauson-Khand reaction, hydroformylation and other methods using metal carbonyl reagents. Palladium-catalysed reactions are not included as this area has been recently reviewed (*vide infra*).

1. INTRODUCTION

Combinatorial chemistry and solid phase synthesis have evolved in the last ten years to become valuable techniques for decreasing the time involved in drug development. To reach its full potential, however, solid phase synthesis has to incorporate the many versatile organometallic reactions developed over recent decades and much progress has indeed been achieved within this area in the last few years. This review covers the literature dealing with organometallic solid phase chemistry from 2000 until 2004, including topics such as Grignard reactions and related reactions, metathesis, hydroformylation and other transformations involving metal carbonyl reagents, Tebbe-type olefinations as well as radical mediated reactions. Some subjects have been excluded however. Palladium-catalysed reactions on solid phase have been recently reviewed by Bräse *et al.* [1], and this area is extensive enough to merit a review on its own. Polymer-bound reagents are not covered, we instead refer to recent reviews by Ley [2, 3], Bradley [4], Leadbeater [5, 6] and Bräse [7]. For earlier treaties about organometallic reactions on solid phase, there are two excellent reviews by Kurth and co-workers concerning carbon-carbon bond formation on solid phase, covering the literature between 1990 and 1999 [8, 9]. Recommended is also a review by Brill [10], which includes experimental procedures for performing organometallic reactions on solid phase, as well as a recent review by Reginato and Taddei [11] covering reactions with Li, Mg, Cu and Zn.

2. GRIGNARD REACTIONS AND RELATED REACTIONS

The Grignard reaction was one of the first organometallic reactions to be used on solid phase and this area has been reviewed by Franzén [12]. O'Donnell *et al.* [13] used a Weinreb-linker in their synthesis of amino ketones and peptide ketones (**1**), where the product was cleaved from the resin using five different Grignard reagents (Scheme 1). All Grignard reagents gave the expected products in good yields, with the exception of *t*-BuMgCl which did not generate any



Scheme 1. Solid phase synthesis of amino ketones and peptide ketones.

product. Alkyl lithium reagents were also tried, but the Grignard reagents in general gave better yields.

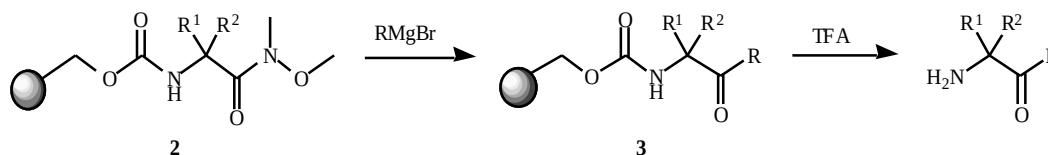
Albericio [14] also used a Weinreb approach for the preparation of ketones on Wang resin. Weinreb amides **2** were treated with three different Grignard reagents to give polymer-bound ketones **3**, which were subsequently cleaved with TFA to liberate the corresponding amines (Scheme 2). A library of 25 compounds was prepared in high yields and purities, incorporating an additional step, i.e. acylation of the amine moiety.

The opposite approach using a commercially available Weinreb amide resin or its benzyl derivative gave low purities of the crude product **4** upon cleavage with Grignard reagents, due to contamination by a cross-coupled byproduct (Scheme 3) [15].

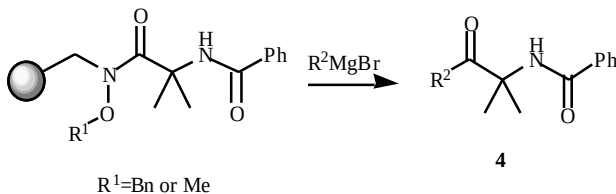
Lazny and Nodzevska [16] compared six different triazene linkers in a Grignard addition to polymer-bound nortropinone (**5**, Scheme 4). Both *meta* and *para*-phenol triazenes were used in combination with two different spacer lengths (3 or 6 carbon diethers) and compared to two earlier reported triazene linkers, i.e. *meta* and *para*-T2 [17]. Best results were obtained with the three-carbon spacer, with yields of 69% for the *meta*- and 71% for the *para*-substitution pattern.

Timmermann *et al.* [18] used 4-iodoimidazole immobilised on 2-chlorotrityl resin (**6**) in a metal/halogen exchange, followed by the addition of different electrophiles (Scheme 5). The substituted imidazole **7** was cleaved from the resin with TFA to afford ketones, thioketones or alcohols in moderate to good yields and purities. Best results were obtained with aldehydes as electrophiles; other electrophiles examined were acid chlorides, nitriles, isocyanates and

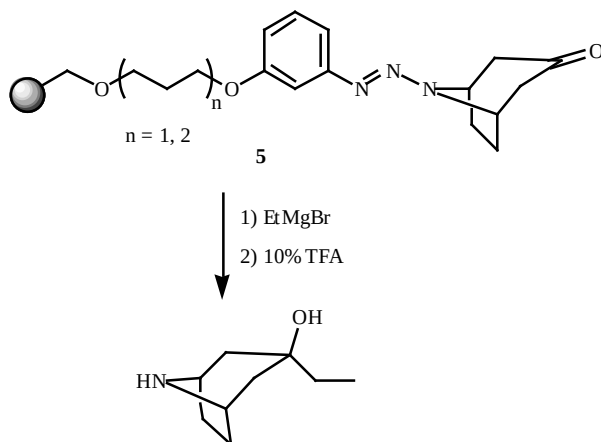
* Address correspondence to this author at Department of Chemical and Biological Engineering, Chalmers University of Technology, SE-41296 Göteborg, Sweden; Tel: +46-31-7723070; Fax: +46-31-7723657; E-mail: kann@chembio.chalmers.se



Scheme 2. Preparation of amino ketones via reaction of Grignard reagents with a polymer-bound Weinreb amide.



Scheme 3. Alternative approach towards aminoketones.



Scheme 4. Solid phase synthesis of nortropinone derivatives.

ketones. For $X = O$, the ketones were reacted further with methyl-, vinyl- or allyl magnesium bromide to give the corresponding tertiary alcohols **8** with satisfactory results.

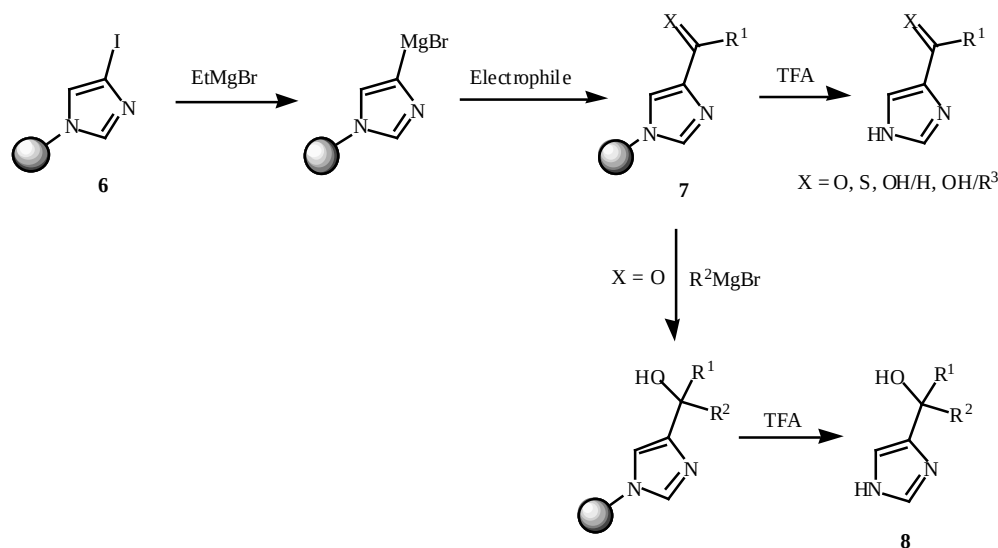
Grigg and co-workers [19] reported the use of Grignard reagents to afford tertiary methylamines. Their strategy,

highlighted in Scheme 6, is based on the formation of solid phase-bound benzotriazole adducts **9**, which undergo a Grignard displacement to give the polymer-bound secondary hydroxylamines **10** [20, 21]. Five different Grignard reagents were used to afford the corresponding tertiary methylamines, generally in good yields although the sterically hindered isopropyl magnesium halide gave the lowest yields (18 and 25% for the two entries using this reagent). This method was also used to synthesise the MAO-inhibitor α -methylpargyline, as well as the analgesic ($\pm$)-Tramadol, on solid phase [22].

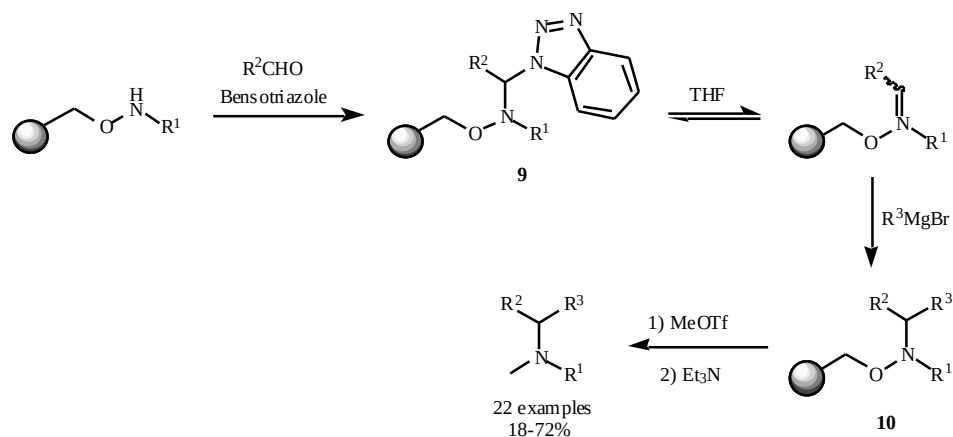
2,3-Disubstituted benzopyran-4-ones were prepared by Showalter and Harikrishnan [23] using a traceless silyl linker (Scheme 7). The polymer-bound aryl aldehyde **11** participated in a Grignard reaction with three different benzyl magnesium chlorides, followed by an IBX (1-hydroxy-1,2-benziodoxol-3(1H)-one) assisted oxidation to form ketones **12**. After a few more reaction steps on solid phase, the benzopyranones **13** were liberated with cesium fluoride to give the desired products in moderate to good yields and in excellent purities.

Hanquet and Solladié [24] describe the use of menthyl sulphinate **14** (racemic on sulphur) in the preparation of polymer-bound sulfoxides **15** (Scheme 8). **14** was reacted with four different Grignard reagents (methyl, ethyl, phenyl and benzyl) to give quantitative yields in all four cases, apart from the benzylic Grignard reagent which gave a yield of 75%.

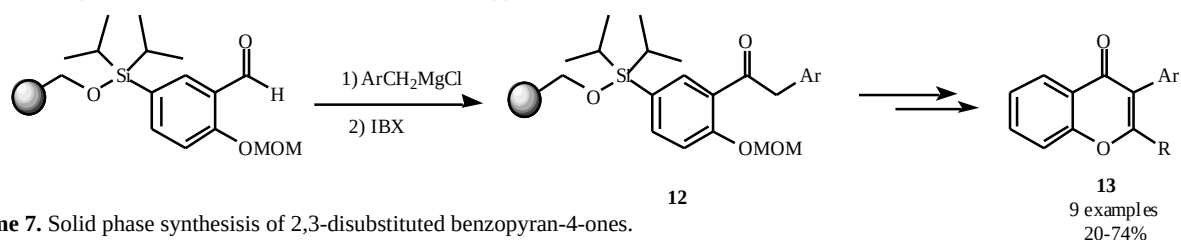
In the solid phase synthesis of dysidiolide analogues (Scheme 9), Waldmann *et al.* [25] used 3-bromomethylfuran and magnesium to incorporate a furan moiety into a polymer-bound aldehyde scaffold (**16**), followed by the



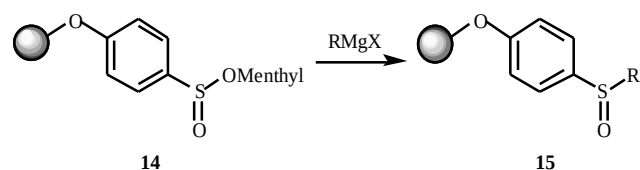
Scheme 5. Solid phase synthesis of 4-substituted imidazoles via polymer-bound Grignard reagents.



Scheme 6. Preparation of tertiary methylamines on solid support.



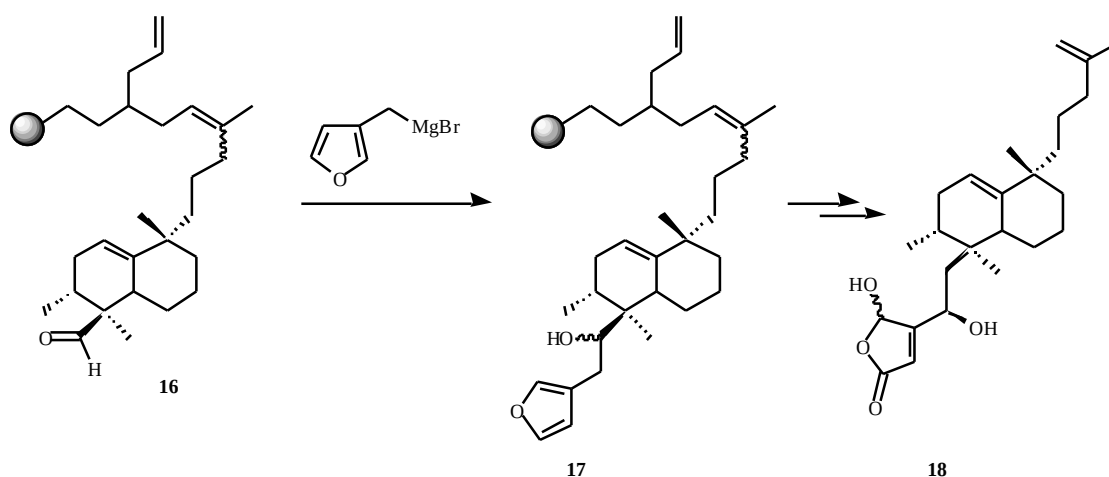
Scheme 7. Solid phase synthesis of 2,3-disubstituted benzopyran-4-ones.



Scheme 8. Preparation of polymer-bound sulfoxides.

metal-mediated reactions including treatment with Grignard reagents. Polymer-bound methyl ester **19** was treated with a large excess of allyl magnesium bromide at elevated temperature yielding the desired alcohols **20** in good yield after an oxidative cleavage.

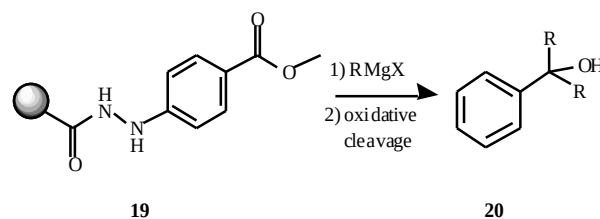
Bräse [27] and co-workers synthesised phthalides, i.e. benzobutyrolactones, using a Grignard or dialkylzinc



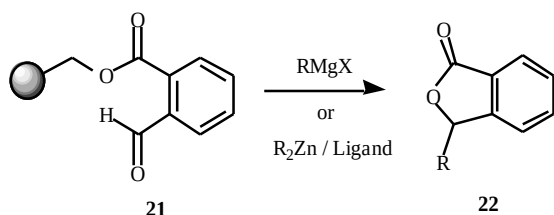
Scheme 9. Solid phase synthesis of dysidiolide analogues.

oxidation of alcohol **17** to hydroxy lactone **18** after cleavage from the polymer *via* a ruthenium-catalysed ring-closing metathesis (see also section 6). Only a low degree of conversion was obtained in the Grignard reaction and the yields could not be improved by using an excess of reagent or raising the temperature.

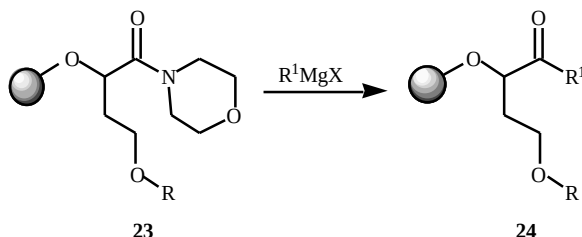
The same research group also developed a traceless phenylhydrazine linker for solid phase synthesis (Scheme **10**) [26]. The acyl hydrazide linker is stable to a variety of



Scheme 10. Preparation of polymer-bound sulfoxides.



Scheme 11. Solid phase synthesis of phthalides using Grignard or zinc reagents.



Scheme 12. Conversion of morpholine amides to ketones using Grignard reagents.

mediated cyclisation in the cleavage step. When polymer-bound aldehyde **21** was treated with Grignard reagents at low temperature (-70°C), the liberated phthalide **22** was obtained (Scheme 11). Use of ethyl magnesium chloride gave predominantly the corresponding benzylic alcohol instead. Higher reaction temperatures gave the tertiary alcohol resulting from addition of the Grignard reagent to the ester functionality. Dialkylzinc reagents were also used in conjunction with chiral ligands, although the enantioselectivity was low in this case.

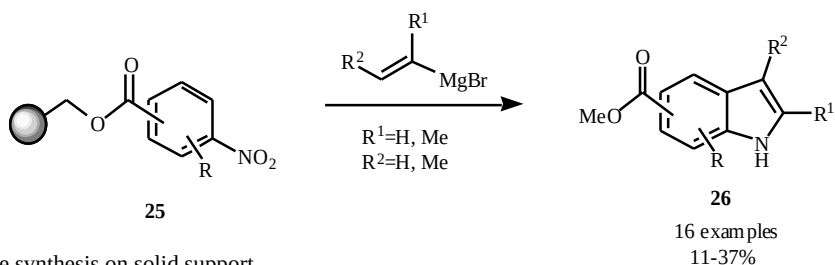
In the development of a samarium-mediated traceless cleavage strategy, further described in section 7, Procter *et al.* [28] used polymer-bound morpholine amide **23** as the electrophile in the addition of Grignard reagents to form the corresponding ketones **24** (Scheme 12).

Bräse and Knepper [29] used a Bartoli indole protocol on Merrifield resin to obtain tetrasubstituted indoles. Polymer-bound aryl nitro ester **25** was treated with different vinyl magnesium bromides, and the heterocyclic moieties were subsequently liberated with sodium methoxide in THF to furnish indoles **26** in low yields but good to excellent purities (Scheme 13). The lower yields can be explained by cleavage from the resin during the Grignard addition reaction.

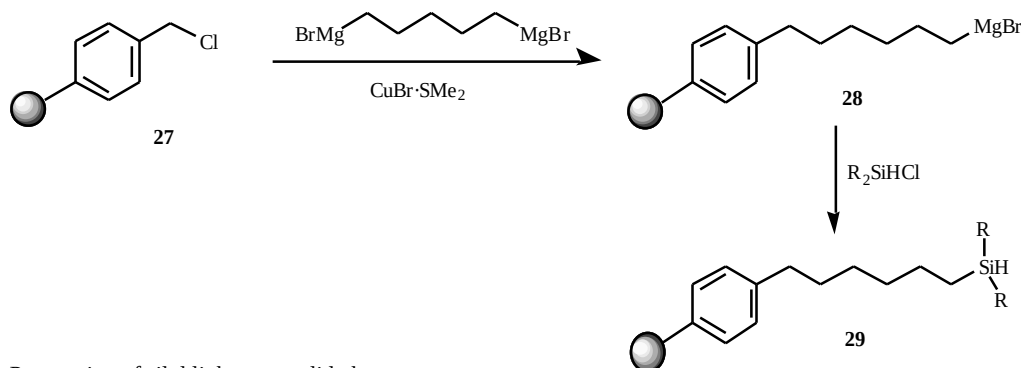
Takahashi *et al.* [30] use a di-Grignard reagent for the synthesis of various silyl linkers. Two linkers were investigated for the attachment of alcohols. Polymer-bound benzyl chloride **27** (Merrifield, Argogel, Wang, Argogel Wang) was treated with an excess of pentane-1,5-di(magnesium bromide) in the presence of copper(I)bromide to give the cross-coupled polymer-bound Grignard reagent **28** after removal of excess Grignard reagent by filtration (Scheme 14). **28** was then treated with two different chlorosilanes (diphenyl and diisopropyl) to furnish polymer-bound silane **29** as indicated by the IR absorption at $2088\text{--}2094\text{ cm}^{-1}$. The loading was determined by chlorination and reaction with a Fmoc-protected amino acid, and a subsequent Fmoc cleavage test [31], giving yields in the range of 51 to 79%.

Chang and Kurth [32] applied a sulphone linker strategy in the synthesis of 3,5-disubstituted cyclopent-2-enones. Polymer-bound epoxide **30** was reacted with a Grignard reagent (allyl magnesium bromide, benzyl magnesium bromide) or a cuprate ($n\text{-Bu}_2\text{Cu}(\text{CN})\text{Li}_2$) to form alcohol **31** (Scheme 15). The alcohol was then oxidised using either Swern conditions or TPAP/NMO, followed by a base-induced β -elimination yielding the desired disubstituted pentenones **32**.

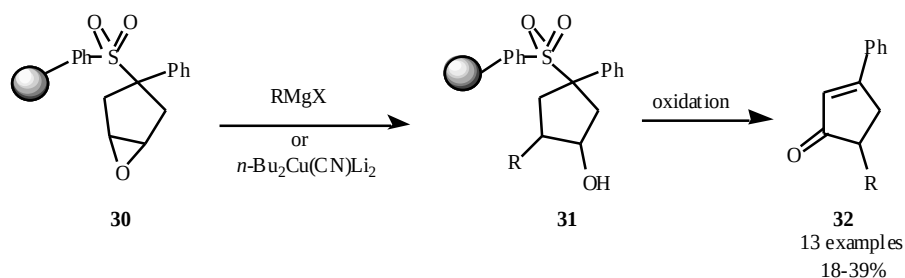
Zech and Kurth [33] treated polymer-bound enaminones (**33**), silyl-linked *via* a galactose auxiliary, with two different cuprates in the presence of a Lewis acid to give *N*-galactosylated 2,6-disubstituted piperidinones (**34**) in high



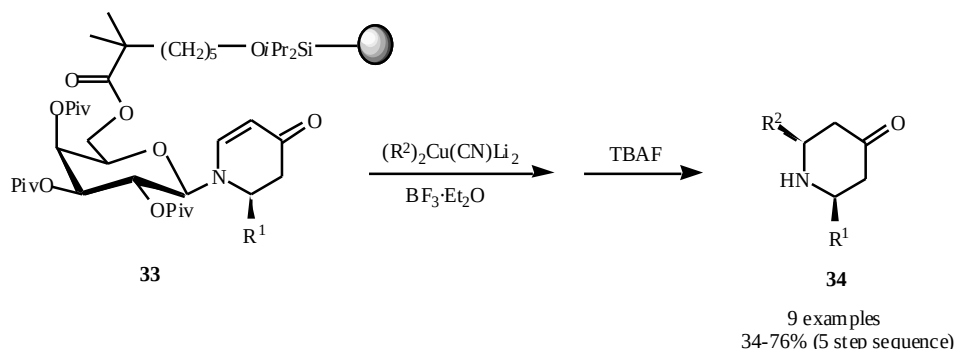
Scheme 13. Bartoli indole synthesis on solid support.



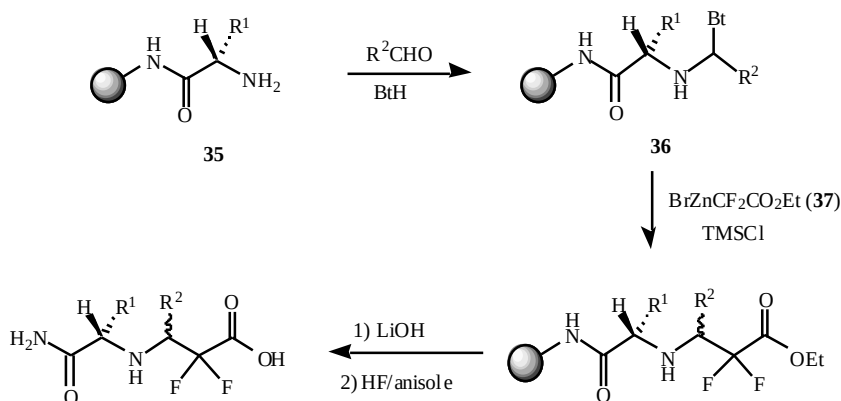
Scheme 14. Preparation of silyl linkers on solid phase.



Scheme 15. Preparation of cyclopentenones on solid phase using a sulphone linker.



Scheme 16. Stereoselective synthesis of piperidinones using a galactose auxiliary.



Scheme 17. Synthesis of α, α -difluorinated- β -amino acids via a Reformatsky procedure.

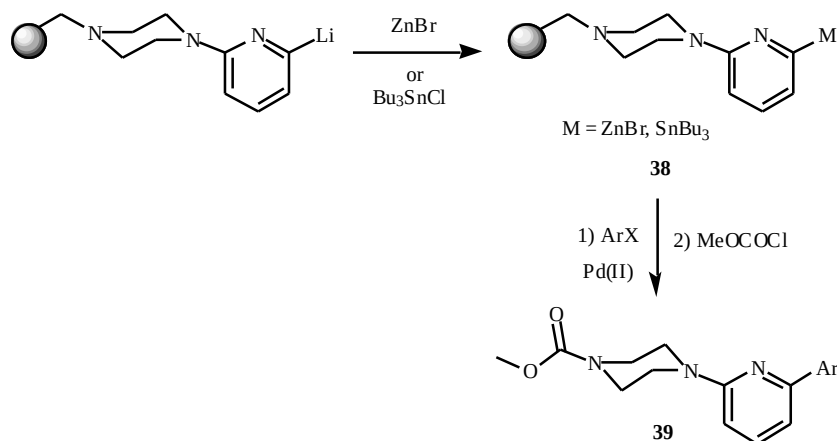
overall yield and excellent *cis* selectivity after cleavage with TBAF (Scheme 16).

Houghten *et al.* [34] used a Reformatsky reaction [35] to synthesise difluoro- β -amino acids on solid phase (Scheme 17). The polymer-bound amino acid **35** was reacted with different aldehydes in the presence of benzotriazole according to a procedure developed by Katritzky [20, 21]. The benzotriazole derivative **36** was then treated with a large excess of the Reformatsky reagent **37** in refluxing THF. Subsequent hydrolysis, followed by treatment with HF, gave the α, α -difluorinated- β -amino acids as diastereomeric mixtures in excellent yields and purities. The use of the corresponding imine as electrophile gave poor results.

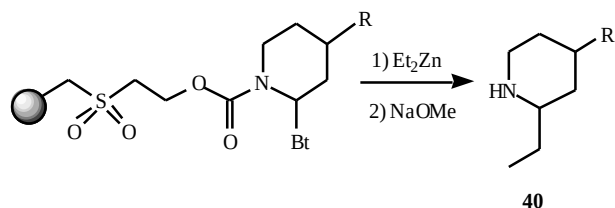
Although, Negishi, Kumada and Stille cross-coupling reactions are usually palladium catalysed, a few examples of these reactions are included in this review due to the nature of the nucleophile. The first example is the reaction of a polymer-bound 2-lithiopyridine which underwent a metal-metal exchange with either zinc bromide or tributyltin

chloride to form the polymer-bound organometallic species **38** (Scheme 18), as reported by Gros and Fort [36]. **38** was subsequently reacted with different aryl or heteroaryl halides to give the corresponding biarylic compounds **39** after cleavage with methyl chloroformate. In the Negishi coupling, phenyl iodide gave an overall yield of 40%, whereas 2-bromopyridine only gave traces of the desired product. In the Stille coupling, three different aryl halides were used (phenyl bromide 34%, phenyl iodide 50%, while 2-bromopyridine gave traces of product).

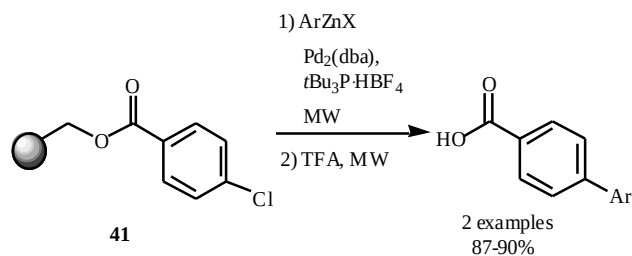
Hiemstra *et al.* [37] used Katritzky's protocol [20, 21] to generate an *N*-acyliminium ion from the corresponding polymer-bound benzotriazole derivative, which was then treated with various carbon nucleophiles to give disubstituted piperidines **40**. Diethylzinc was the the only organometallic nucleophile giving a yield of 14% over four reaction steps (Scheme 19). The addition reaction failed when allenyl tributyltin was used as the nucleophile.



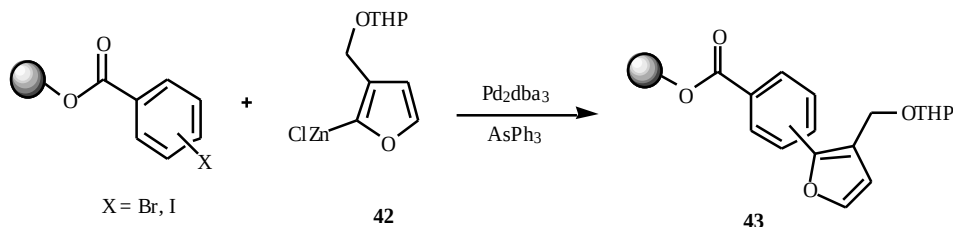
Scheme 18. Preparation of aryl-substituted pyridines using Stille and Negishi couplings.



Scheme 19. Piperidine derivatisation using organozinc reagents.

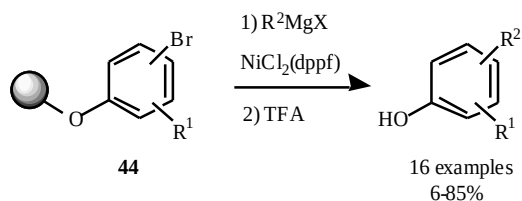


Scheme 20. Microwave assisted synthesis of biaryl compounds.



Scheme 21. Solid phase synthesis of furan derivatives.

Walla and Kappe [38] treated polymer-bound aryl chloride **41** with two different aryl zinc halides (phenyl zinc chloride and 3-methoxyphenyl zinc iodide) under Negishi-



Scheme 22. Nickel-catalysed cross-coupling on solid phase.

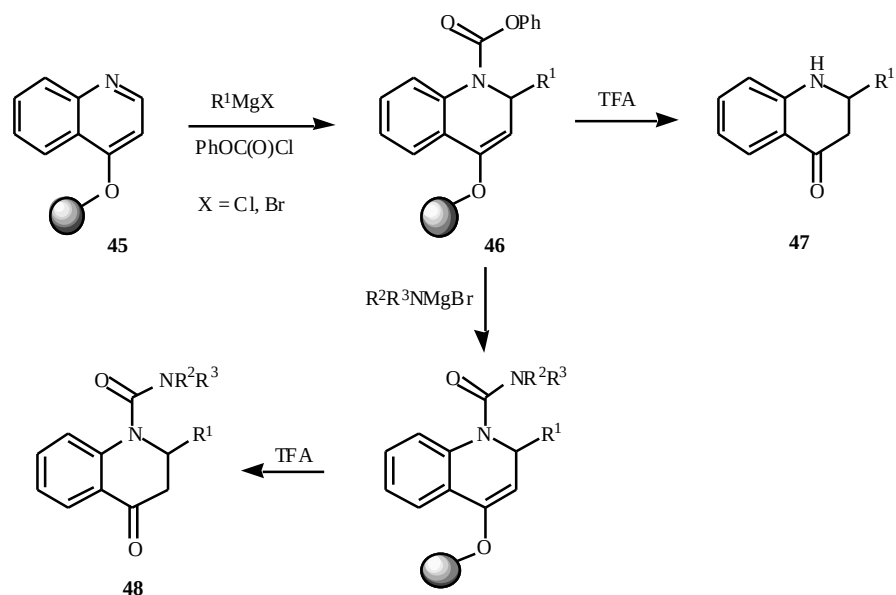
type conditions in a microwave reactor to give the corresponding biaryls in high yields and high purities (Scheme 20) with an overall reaction time of 20 minutes.

Han and co-workers [39] used the zincate **42** as a scaffold for the generation of substituted biaryl furan derivatives on a Wang resin (Scheme 21). **42** is easily prepared *via* lithiation of a THP-protected 3-furan methanol followed by treatment with ZnCl₂ to give the indicated organometallic reagent (contaminated with 5% of the 5-isomer). The zincate was reacted with three different Wang resin-bound aryl halides, forming substituted biaryl furan derivatives **43**.

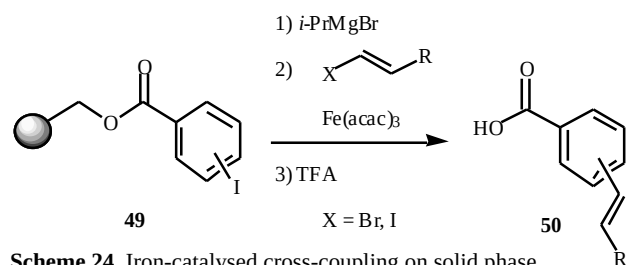
The use of a nickel-catalysed Kumada-type reaction is shown in Scheme 22. Aryl bromides were attached to Wang resin using a Mitsunobu protocol as reported by Lee and co-workers [40]. The polymer-bound aryl bromides **44** were reacted with 10 equivalents of four different Grignard reagents using a Ni(0)-catalyst. All reactions were successful with yields ranging from moderate to high, and in low to excellent purities. *n*-Butylmagnesium chloride showed lower

reactivity compared to the other Grignard reagents, and steric hindrance in the electrophiles was shown to be detrimental for the reaction.

Wendeborn [41] developed a solid phase route to 2,3-dihydroquinoline-4-ones as shown in Scheme 23. Resin-bound quinoline **45**, attached *via* a 1,6-hexanediol spacer to Merrifield resin, was treated with 13 different Grignard reagents in the presence of phenyl chloroformate at -70 °C to obtain the polymer-supported 2-substituted carbamate **46**. Cleavage of **46** with TFA gave 2-substituted-2,3-dihydroquinoline-4-ones (**47**) in excellent yields and purities, only the *iso*-butylmagnesium bromide gave a low yield



Scheme 23. Solid phase synthesis of substituted 2,3-dihydroquinolin-4-ones.



Scheme 24. Iron-catalysed cross-coupling on solid phase.

(57%). In one example, carbamate **46** was reacted further in the sequence and treated with magnesium salts of 10 different amines generated from the reaction with ethylmagnesium bromide, yielding urea derivatives **48** in high yields and excellent purities after cleavage with TFA.

The mild nature of iron-catalysed coupling conditions enables several sensitive functional groups like esters or nitriles to be present under the reaction conditions. Knochel *et al.* [42] describe the addition of a Grignard reagent to a polymer-bound aryl iodide as highlighted in Scheme 24. *Ortho*- and *para*-substituted iodobenzoic acids (**49**), immobilized on Wang resin, were treated with isopropyl magnesium bromide at -20°C to give the corresponding polymer-bound Grignard reagent, which was treated with a large excess of vinyl halide in the presence of a catalytic amount of $\text{Fe}(\text{acac})_3$. The cross-coupled products were then released from the polymer using TFA to give benzoic acid derivatives (**50**) in excellent yields.

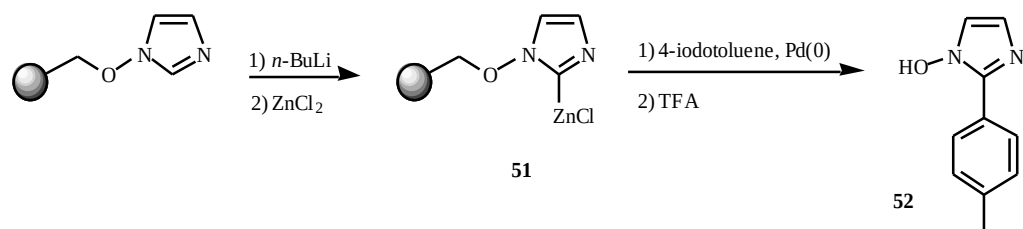
Begtrup and co-workers [43] generated a polymer-bound imidazol-2-ylzinc chloride *via* a metal-metal exchange reaction as shown in Scheme 25. The polymer-bound organozinc reagent **51** was treated under Negishi type conditions with 4-iodotoluene to give 2-tolyl-1-hydroxyimidazole (**52**) in 85% yield after cleavage with TFA. Eight more aryl iodides are exemplified using the same protocol.

A polymer-bound cyclic bismethylthiomethylene malonic ester (**53**) was diversified using Grignard reagents in the synthesis of substituted quinolones (Scheme 26), as reported by Tang and Huang [44].

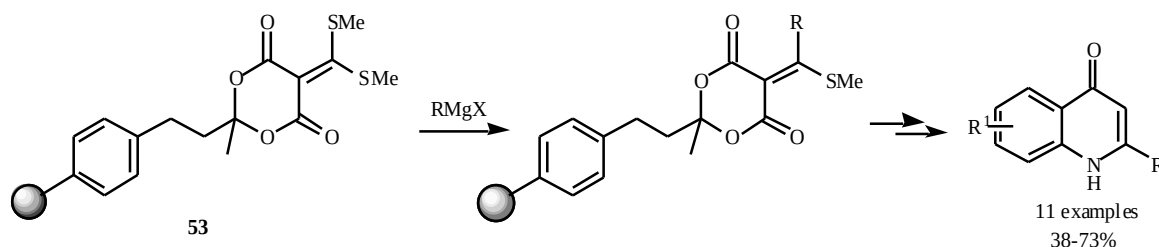
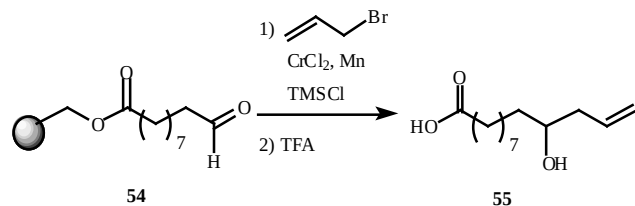
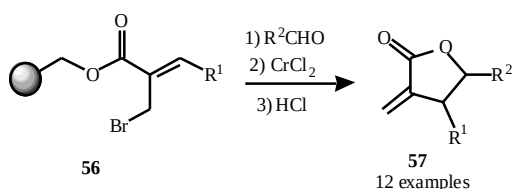
Hari and Miller [45] treated the polymer-bound aldehyde **54** with an excess of allyl bromide in the presence of sub-stoichiometric amounts of CrCl_2 , excess manganese and excess TMSCl to give the allylic alcohol **55** (Scheme 27) in an overall yield of 72% (4 steps). The elegance in this Nozaki-Hiyama-Kishi coupling [46-48] is the use of a frit for separating the manganese from the polymer, allowing the reducing agent to be regenerated and also facilitating the washing of the resin.

An inverted approach of the Nozaki-Hiyama-Kishi coupling was used by Delgado *et al.* [49]. Polymer-bound allyl bromide **56** was treated with different aldehydes in the presence of CrCl_2 to give lactones **57** in moderate yields after cleavage with HCl in THF (Scheme 28).

Wessjohann *et al.* [50] studied homogeneous chromium-mediated solid phase reactions (Reformatsky and Takai-



Scheme 25. Solid phase synthesis of substituted 1-hydroxyimidazoles.

**Scheme 26.** Solid phase synthesis of quinolone-derivatives.**Scheme 27.** Lactone synthesis on solid phase.**Scheme 28.** Allylation of an aldehyde using a chromium reagent.

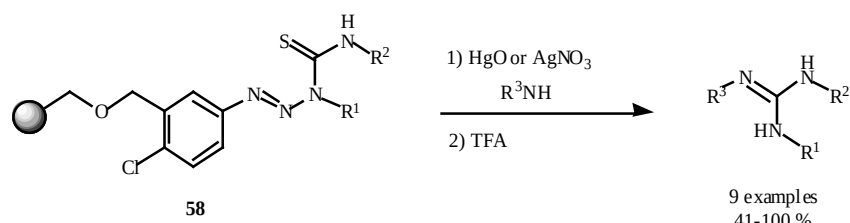
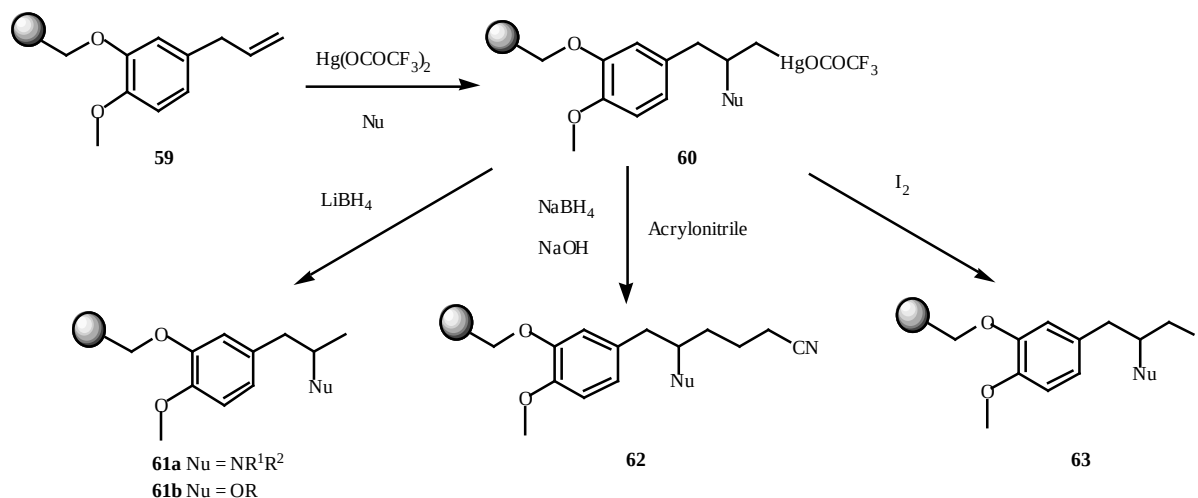
Utimoto reactions) towards an iterative polyol strategy. In their studies they found some limitations of the chromium-mediated solid phase reactions. These reaction types are severely sensitive to the polymer backbone, spacer and linker type, mainly due to the polarity as well as complexation

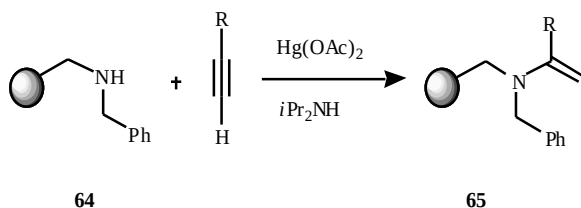
properties of the polymer backbone and the spacer. These reactions are also sensitive to steric hindrance, and a separation of the substrate from the polymer backbone via a polyethylene glycol spacer resulted in an increased yield.

3. MERCURY-PROMOTED REACTIONS

Bräse and Dahmen [51] used a triazene based linker T2* to synthesise trisubstituted triazenes via a mercury(II)oxide promoted transformation of thioureas to the corresponding guanidines. The polymer-bound thiourea **58** in Scheme 29 was treated with three different amines (ammonia, allylamine and piperidine) in the presence of HgO to give the trisubstituted guanidines as their TFA salts after cleavage. The guanidines were formed in good to excellent yield and in high purity. However, allylamine was not compatible with HgO but the mercury salt could be replaced by silver nitrate or circumvented using a tosyl chloride protocol.

The use of alkoxy- and amino-mercuration of alkenes, reported by Raghavan *et al.* [52], is highlighted in Scheme 30. Wang resin-bound eugenol (**59**) was reacted with an excess of amine or alcohol in the presence of mercury(II)trifluoro acetate to furnish the polymer-bound adduct **60**, as reported by Raghavan *et al.* When amines were

**Scheme 29.** Solid phase synthesis of guanidine-derivatives.**Scheme 30.** Alkoxy- and amino-mercuration on solid phase.



Scheme 31. Enamine synthesis using mercury(II) acetate.

used as nucleophiles, the organometallic species was treated either with LiBH_4 to afford amine **61a**, or subjected to a Giese protocol [53] to give the aminonitrile **62**. The alkoxy-derivative of **60** was also treated with LiBH_4 to give **61b**, while adding iodine gave the alkoxyalkyl halide **63**. Similar strategies were also successfully employed using a polymer-bound allyl ester or a polymer-bound cinnamate ester.

Aznar *et al.* [54] reported that secondary amines immobilised on Wang resin (**64**) could undergo an amination reaction with terminal acetylenes, catalysed by mercury(II) acetate, to give the corresponding substituted enamines (**65**, Scheme 31).

Houghten *et al.* [55] prepared bicyclic guanidines on solid phase (Scheme 32). Polymer-bound (MBHA resin) trityl-protected polyamine **66** was treated with thiocarbonyldiimidazole to give the bicyclic guanidine **67**, where the reaction was driven to completion by the addition of mercury(II) acetate. Eleven different polymer-bound bicyclic guanidine-derivatives were synthesized successfully in good yields and purities.

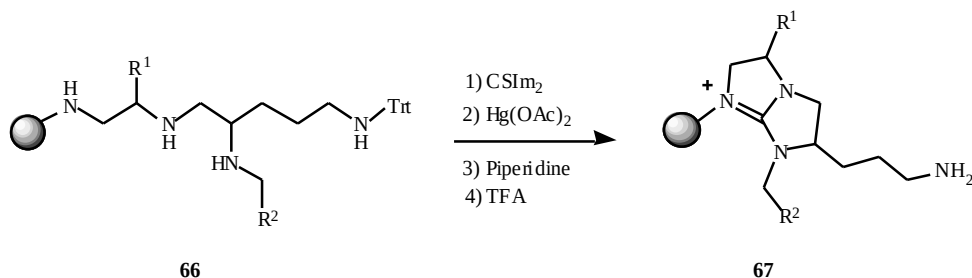
Briehn and Bäuerle [56] iodinated a polymer-bound thiophene (**68**) in a reaction mediated by a mercury(II) salt in their development of a p-conjugated polymer (Scheme 33).

4. TITANIUM ALKYLIDENES

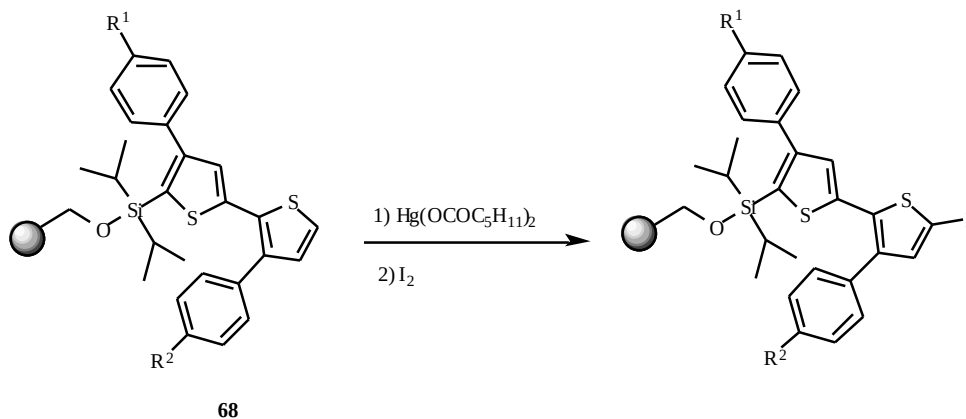
Excluding the use of titanium compounds as Lewis acids, titanium reagents have mainly been applied in the Tebbe olefination and similar reactions on solid phase. Barrett *et al.* [57] have previously reported the conversion of polymer-bound carboxylic esters to vinyl ethers *via* a Tebbe reaction. A further extension of this chemistry involves the solid phase preparation of isoxazoles (**70**) using a combined Tebbe olefination/cycloaddition (Scheme 34, path a) [58]. Further diversity could be introduced *via* a Suzuki reaction on an adjacent aryl halide moiety before the cycloaddition step, forming a small library of eight compounds of high purity (**72**). Direct cleavage of **71** with TFA (path c), excluding the cycloaddition step, gave instead the corresponding ketones (**73**) making this a very versatile reaction sequence.

Hartley *et al.* [59-64] used the Takeda procedure [65, 66] in a related approach towards the traceless solid phase synthesis of benzofurans, indoles, benzothiophenes, and recently also quinoline derivatives. A low-valent titanium complex, $\text{Cp}_2\text{Ti}[\text{P}(\text{OEt})_3]_2$, in combination with a thioacetal containing the desired substituents for incorporation into the olefin, yielded a titanium benzylidene (**74**) containing a masked nucleophile, i.e. O, N or S (Scheme 35). Reaction with a polymer-bound ester produced a supported enol ether (**75**). Deprotection of the nucleophile with TFA, followed by in situ cyclisation and cleavage, gave benzofurans, indoles or benzo[*b*]thiophenes, depending on the identity of the nucleophile. The heterocycles were formed in high purity.

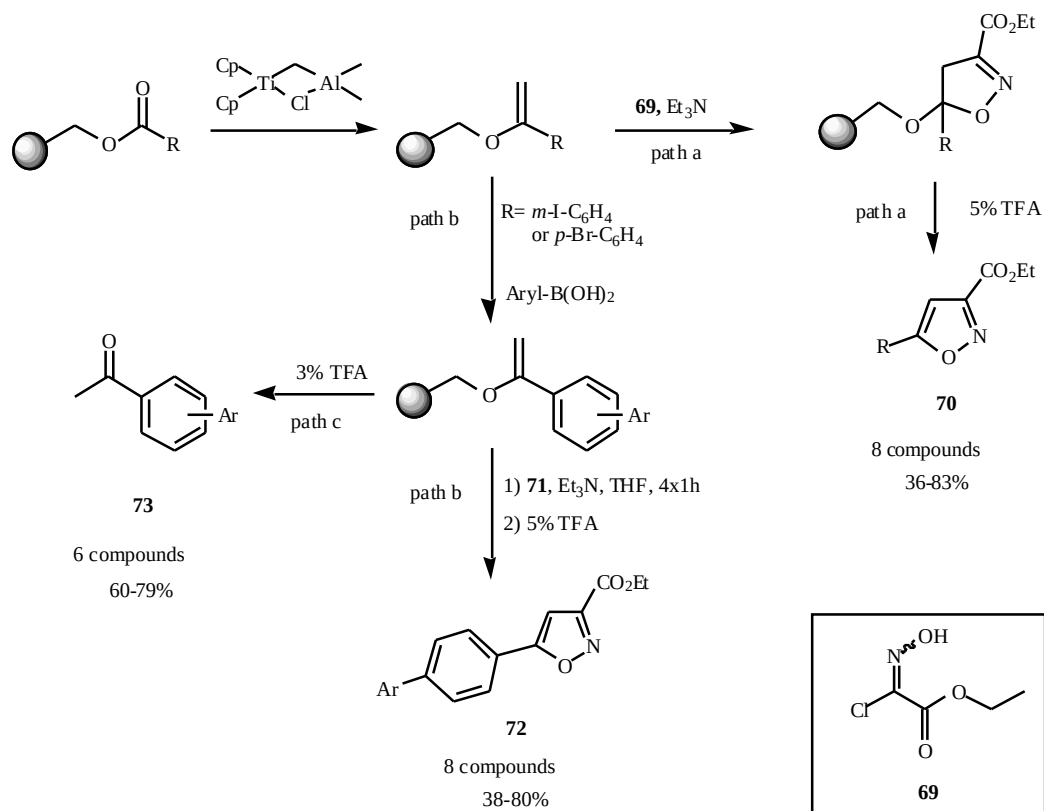
To introduce further diversity in the synthesis of benzofurans, a titanium alkylidene reagent with a pinacolatoboronate moiety incorporated into the structure (**76**) was prepared [61]. This allowed derivatisation of the



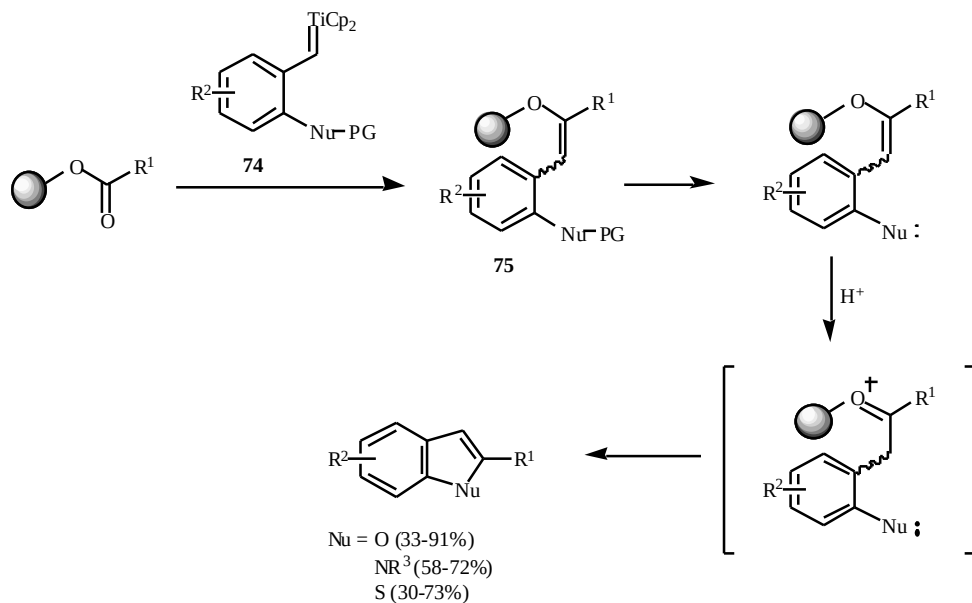
Scheme 32. Preparation of bicyclic guanidines on solid support.



Scheme 33. Mercury-mediated iodination of thiophenes on solid support.



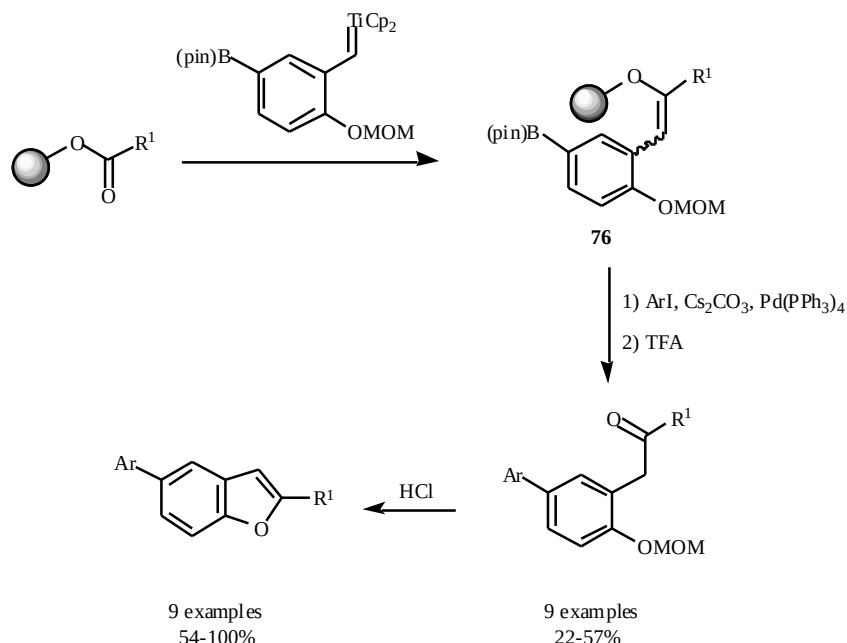
Scheme 34. Solid phase synthesis of isoxazoles and ketones.



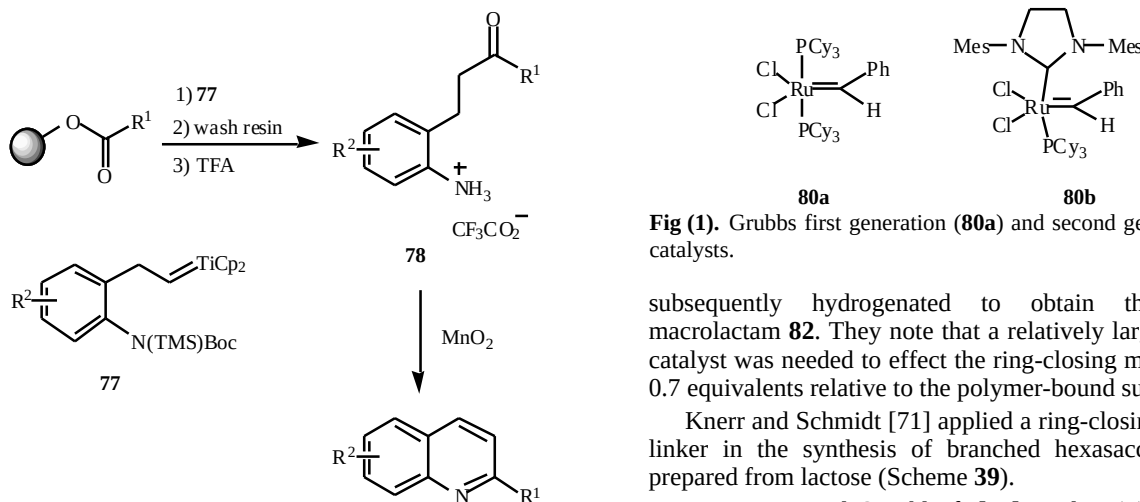
Scheme 35. Solid phase synthesis of benzofurans, indoles and benzo[*b*]thiophenes using titanium benzylidene complexes.

aromatic ring *via* palladium catalysed cross-coupling with aryl iodides before cleavage with TFA (Scheme 36). Cyclisation in this case was effected with 10% HCl in methanol. The intermediate ketones could also be isolated, and two small libraries of 9 compounds each (9 ketones and 9 benzofurans) were prepared.

In the synthesis of quinolines [64], a titanium alkylidene containing an additional methylene group was used to form ketone **78**, which was cyclised and aromatised using manganese dioxide (Scheme 37). Seven different quinolines were prepared in 41-67% yield.



Scheme 36. Synthesis of ketones and benzofurans on solid phase using titanium alkylidene with a boronate moiety incorporated.



Scheme 37. Quinoline synthesis on solid phase using titanium reagents.

5. METATHESIS

The metathesis reaction, i.e. the metal-catalysed coupling of two alkene moieties to form two new alkenes has been known for quite some time, but has seen a renewed interest the last few years due to improvements of the reagents used for this transformation [67, 68]. Most solid phase applications of the metathesis reaction involve the use of ring-closing metathesis (RCM), either for ring formation on the solid support, or as a combined cyclisation/cleavage strategy [69]. Maier and co-workers [70] used the latter strategy to synthesise three different peptidomimetic structures using solid phase methodology (Scheme 38). Diene **79**, prepared using the Evans asymmetric aldol reaction, was cyclised and cleaved in one step using the first generation Grubbs' catalyst **80a** (Fig. (1)), forming macrolactams **81**. In one case, the macrolactam was

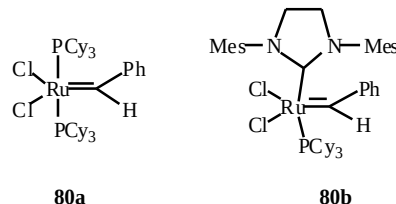


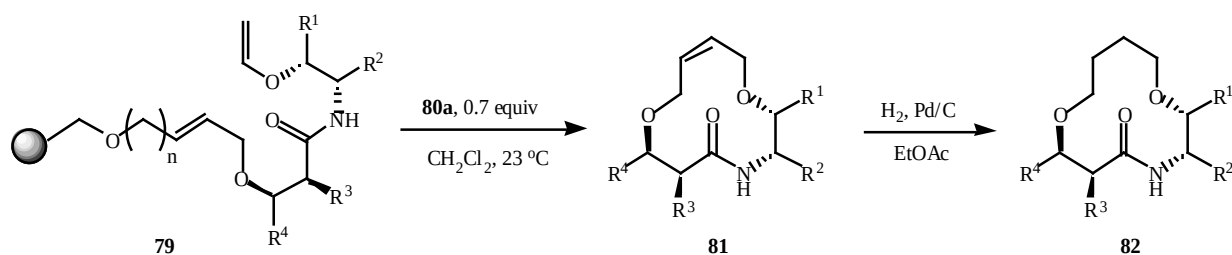
Fig (1). Grubbs first generation (**80a**) and second generation (**80b**) catalysts.

subsequently hydrogenated to obtain the saturated macrolactam **82**. They note that a relatively large amount of catalyst was needed to effect the ring-closing metathesis, i.e. 0.7 equivalents relative to the polymer-bound substrate.

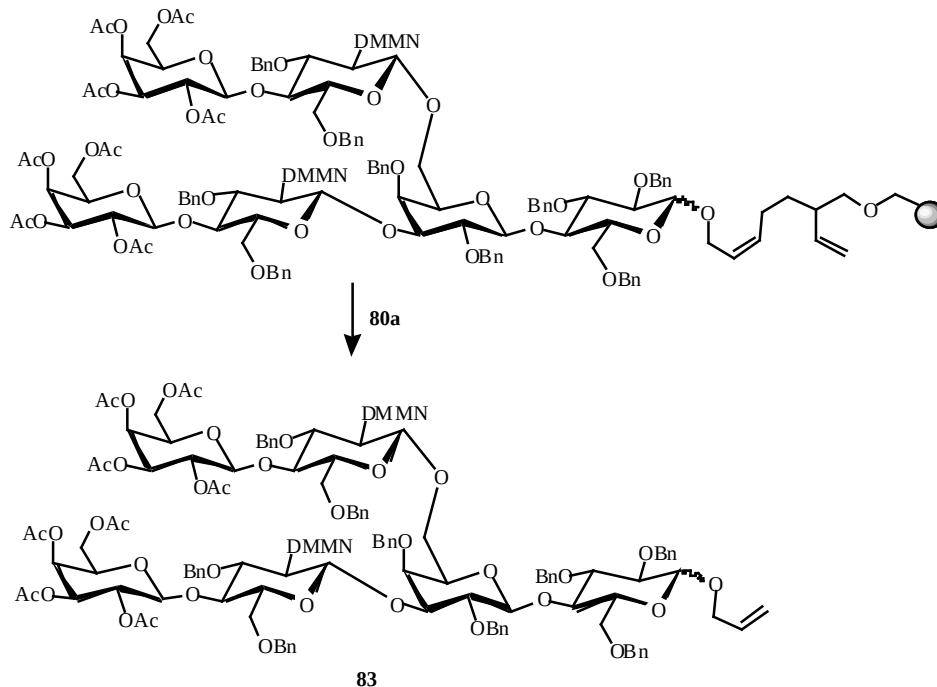
Knerr and Schmidt [71] applied a ring-closing metathesis linker in the synthesis of branched hexasaccharides (**83**) prepared from lactose (Scheme 39).

Van Boom and Overkleeft [72] used D-(+)-mannitol as the basis for a conformationally strained combinatorial scaffold prepared by fusing an oxacyclic ring to the sugar template (Scheme 40). Polymer-bound scaffold **84** was prepared in several steps from D-mannitol, and the unprotected alcohol moiety was treated with three different isocyanates to yield carbamates. Further diversification was effected by reduction of the azide functionality to an amine followed by reaction with acid chlorides. Cyclative cleavage was performed using the second generation Grubbs catalyst **80b**, affording a library of nine different pyranofurans (**85**) in high yields.

Brown and co-workers [73, 74] evaluated the use of a double-armed linker **86b** in the preparation of cyclic sulfonamides, envisaging that the extra linker arm would replace the olefin co-factor that is used to regenerate the catalytic species (Scheme 41). The corresponding single-armed linker **86a** was also prepared for comparison. They found that when using lower amounts of catalyst (< 5%, the double-armed linker was superior to **86a**, affording the



Scheme 38. Solid phase synthesis of peptidomimetics using a metathesis strategy.



Scheme 39. Oligosaccharide synthesis using a RCM linker strategy.

desired cyclic sulfonamides in excellent yields ($\text{R} = \text{Boc}$, methyl, benzyl, 2-bromobenzyl or 2,5-dimethylbenzyl).

A different type of cyclisation-cleavage strategy was used by Waldmann [25, 75] in the synthesis of dysidiolide-derived phosphatase inhibitors (Scheme 42). In this case, the cyclic residue remained attached to the polymer and the desired dysidiolide analogues (**87**), containing a terminal alkene unit, were cleaved in 6–27% yield (8–11 steps).

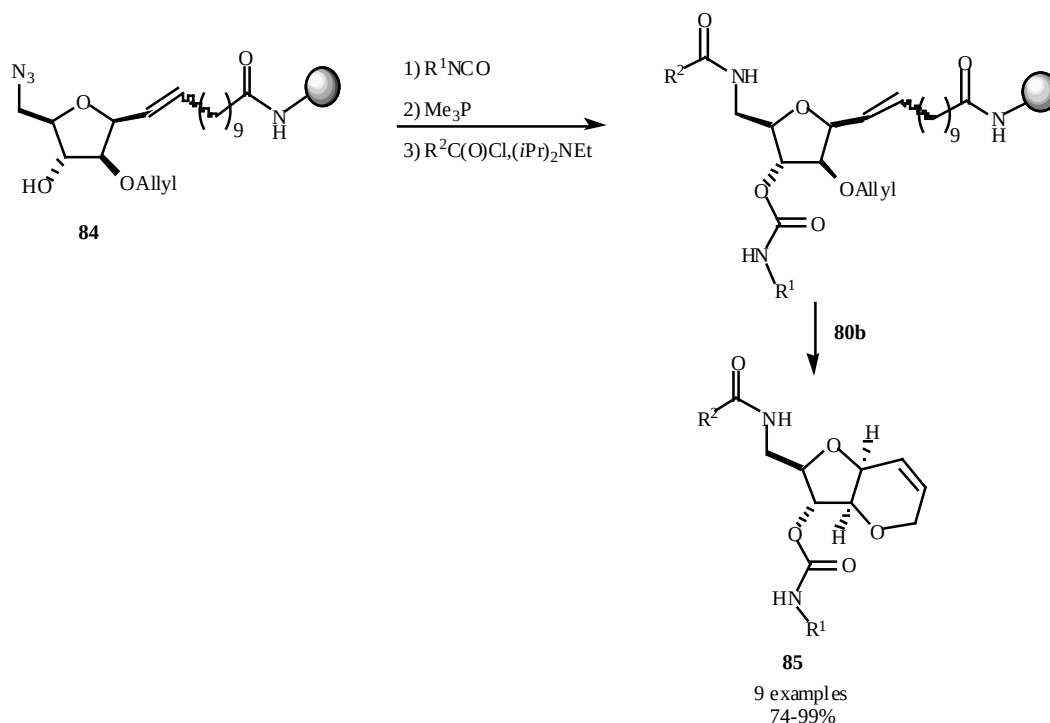
In other cases the cyclic product is formed on solid phase *via* metathesis and the product cleaved from the resin in the conventional way. Arya *et al.* [76–78] have applied RCM in the preparation of tetrahydroquinolines with fused rings of different sizes to produce natural product-like scaffolds for combinatorial synthesis. Scheme 43 shows the preparation of a tetrahydroquinoline structure with a fused 10-membered ring (**88**), but 6- and 8-membered fused rings were also synthesised using the same strategy. Bromo-substituted Wang resin was used as the solid support, and the cyclisations were in this case effected with Grubbs second-generation catalyst **80b**.

Verdine and co-workers [79] have reported the preparation of a library of small-molecule compounds that act as heterodimerisers, based on an invariant ligand moiety diversified with 320 different tetrahydrooxazepines. AP1867,

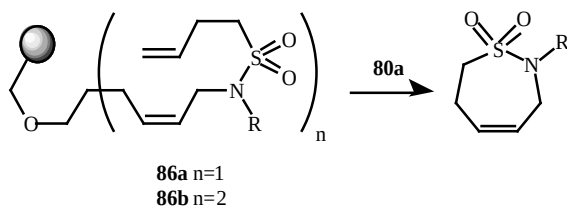
a synthetic analogue of rapamycin and FK506, was chosen as the ligand, while the core structure of the tetrahydrooxazepine was prepared on solid phase involving the use of a ring-closing metathesis reaction (Scheme 44). The cleaved ligands **89** were screened to assess their cell permeability properties.

Ring-closing metathesis has also been used for the cyclisations of peptidic structures on solid support. One interesting examples is that of Amblard and co-workers [80], where rigid bicyclic hexapeptides (**90**) were formed *via* an initial metathesis reaction on solid phase followed by a second cyclative peptide-coupling step in solution (Scheme 45). It is known that conformational restriction of cyclic peptides *via* a second cyclisation step can produce structures with improved binding affinity towards peptide receptors.

Other examples where ring-closing metathesis has been used for the formation of cyclic peptides on solid support include the synthesis of peptide-porphyrin conjugates designed for cancer phototherapy by Krausz and co-workers [81], the preparation of ligands for the study of protein-protein interactions by Schmiedeborg and Kessler [82], as well as a report by Leatherbarrow and co-workers [83] concerning the preparation of serine protease inhibitors.



Scheme 40. Preparation of fused oxacycles from D-(+)-mannitol.

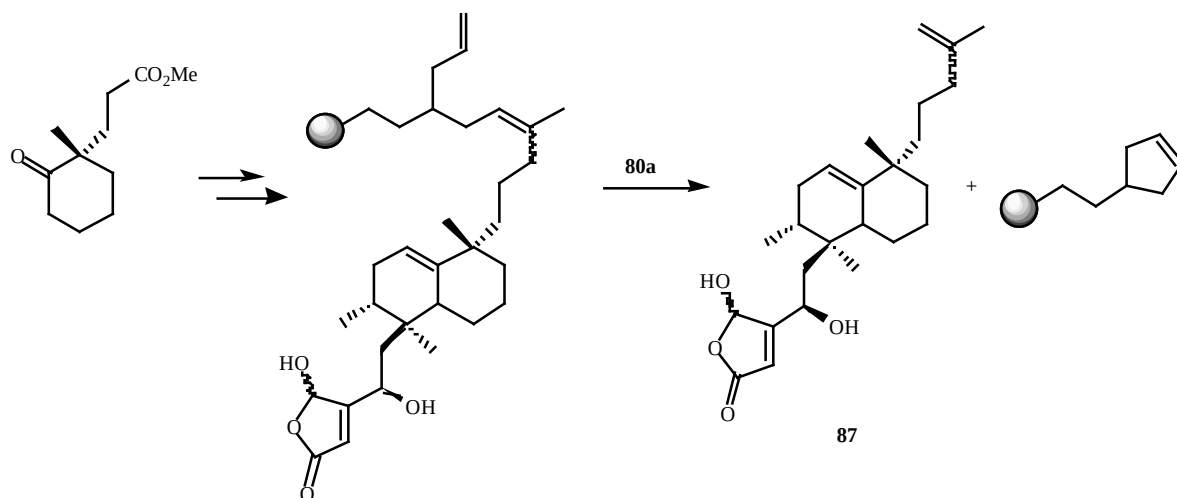


Scheme 41. Solid phase synthesis of cyclic sulfonamides via metathesis.

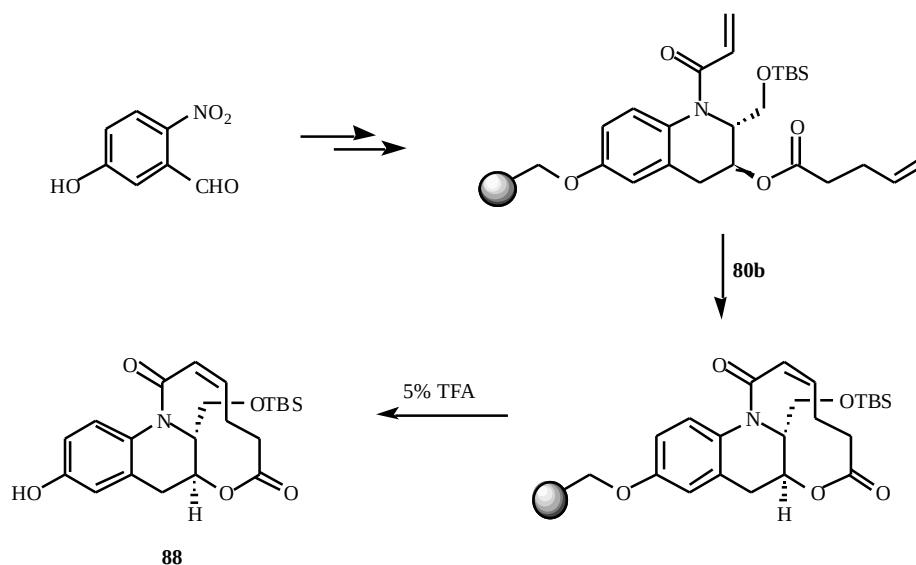
Several groups have studied or exploited site-site interactions on solid support by comparing the extent of intramolecular ring-closing metathesis as compared to intermolecular cross-metathesis. Tang and Wareing [84]

carried out ring-closing metathesis on **91**, forming **92**, noting to their surprise that the extent of cross-metathesis resulting in oligomers was more extensive in the solid phase reaction than in the corresponding reaction in solution, indicating a large amount of site-site interaction between alkene moieties of adjacent polymer-bound molecules (Scheme 46). Several other substrates were also studied, confirming the initial result. Breed *et al.* [85] also report competing interchain metathesis in the cyclisations of polymer-bound allyl substituted amides.

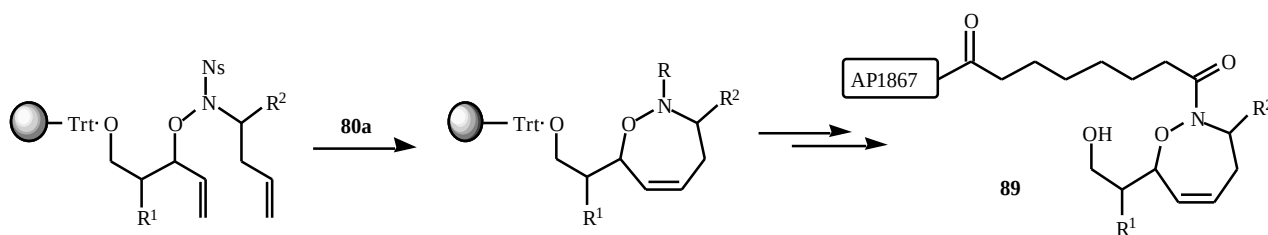
Conde-Frieboes and co-workers [86] have applied ring-closing metathesis in the synthesis of dimeric ligands, where two adjacent alkene moieties react on solid phase but where the final cleaved product has an open-chain structure. This



Scheme 42. Synthesis of dysidiolide analogues using a metathetic cleavage.



Scheme 43. Preparation of a tetrahydroquinoline structure with a fused 10-membered ring.



320 examples

Scheme 44. Library synthesis of cell permeable molecules.

concept was also used by Schreiber and co-workers [87] in the dimerisation of polymer-bound amino acids with alkene-containing side-chains of different length. A similar approach was used by Dervan and co-workers [88], where chains of polyamides containing alkene-linkers of different lengths, were cross-linked *via* a metathesis reaction, yielding dimeric bridged structures after cleavage (Fig. (2)).

Examples of olefin metathesis on solid phase, i.e. not involving ring-closure, are fewer. Seeberger *et al.* [89-91] have developed an oct-4-ene-1,8-diol linker (**93**) for solid phase oligosaccharide synthesis that used olefin metathesis with ethylene to achieve cleavage of the final product. The method was applied in the solid phase synthesis of a potential vaccine for the tropical disease leishmaniasis, where the vaccine is based on a branched tetrasaccharide structure (Scheme 47). The synthesis was performed both in a non-automated and in an automated fashion.

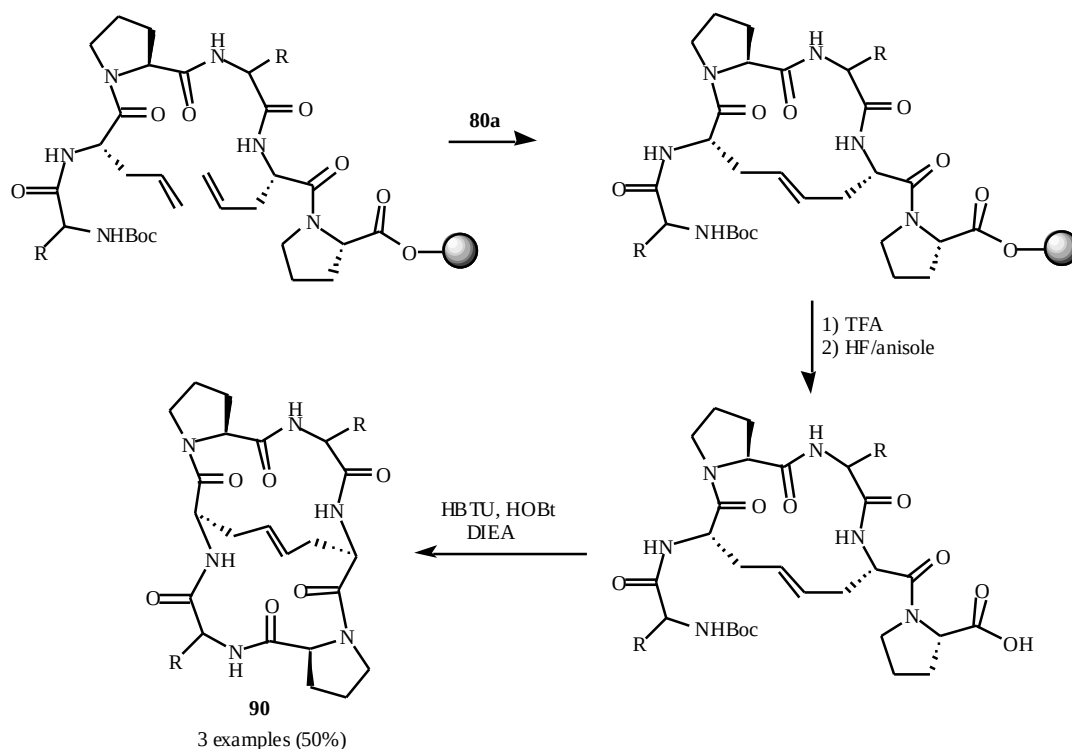
Other applications include the automated synthesis of a potential malaria vaccine with a hexasaccharide structure [92], and the preparation of the proteoglycan-linkage [93]. Four different resins were screened for use in the automated solid phase oligosaccharide synthesis, i.e. Tentagel™, Argogel™, Argopore™ and Merrifield resin, with the latter giving the best results in most cases. A masked alkene linker in the form of a dibrominated linker, which could be reductively debrominated to an alkene before cleavage by metathesis, was also developed [94].

Chang *et al.* [95] used solid phase cross-metathesis for the preparation of unsymmetrical stilbenoid derivatives, thus avoiding the formation of self-coupled stilbenes which can be problematic in solution. 4-Vinylphenol was attached to Merrifield resin and coupled to different styrene derivatives using Grubbs second generation catalyst, yielding stilbenes **97** in 54-72% yield after cleavage with TFA (Scheme 48).

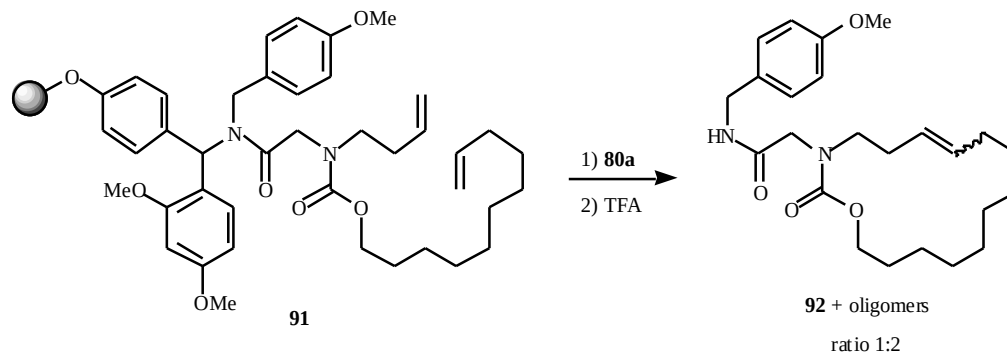
Cross-metathesis on solid phase has also been used by Liao [96] in the synthesis of symmetrical dimeric benzo[b]furanes, and by Verdine [97] in the preparation of opioid-receptor ligands.

6. RHODIUM-CATALYSED CARBENE INSERTION

Iso *et al.* [98] prepared resin-bound α -diazoketone **98**, as well as an aliphatic analogue, from polymer-supported acid chlorides. The α -TMS-diazoketones could be stored at room temperature for at least a month, if protected from air and light. To demonstrate the applicability of the α -TMS-diazoketones in synthesis, **98** was treated with rhodium(II) acetate and thiophenol or aniline, causing insertion of the carbenoid species into the H-S or H-N bond, yielding α -substituted ketones (**99**) after cleavage with TFA (Scheme 50). Oxazoles (**100**) were also prepared on solid phase by conversion of the aromatic α -TMSdiazoketone to an ylide, followed by reaction with aryl nitriles. Similar reactions with the corresponding aliphatic substrate were also carried out successfully.



Scheme 45. Preparation of rigid bicyclic hexapeptides on solid phase.



Scheme 46. Ring-closing metathesis on solid support.

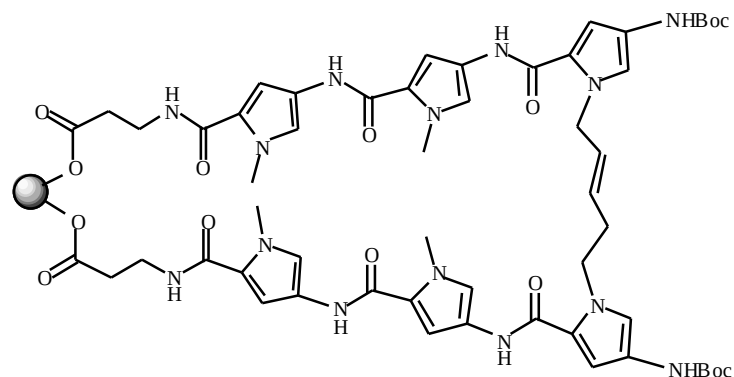
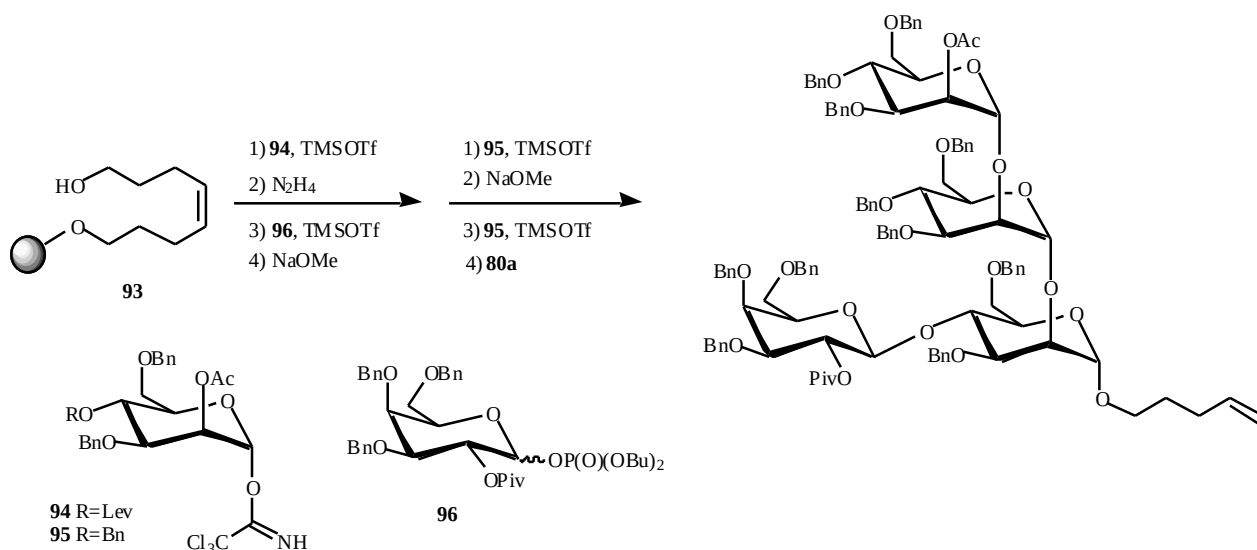


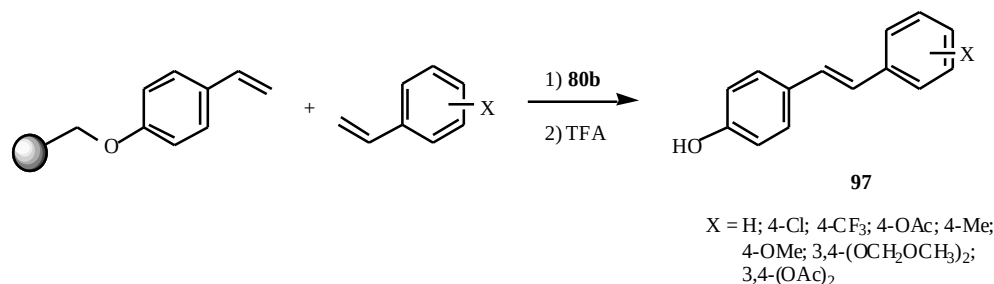
Fig. (2). Dimerisation of polymer-bound polyamides via cross-metathesis.

A solid phase intramolecular Buchner reaction is also reported, yielding 7-hydroxy-2-tetralone (**101**) after cleavage (Scheme 50).

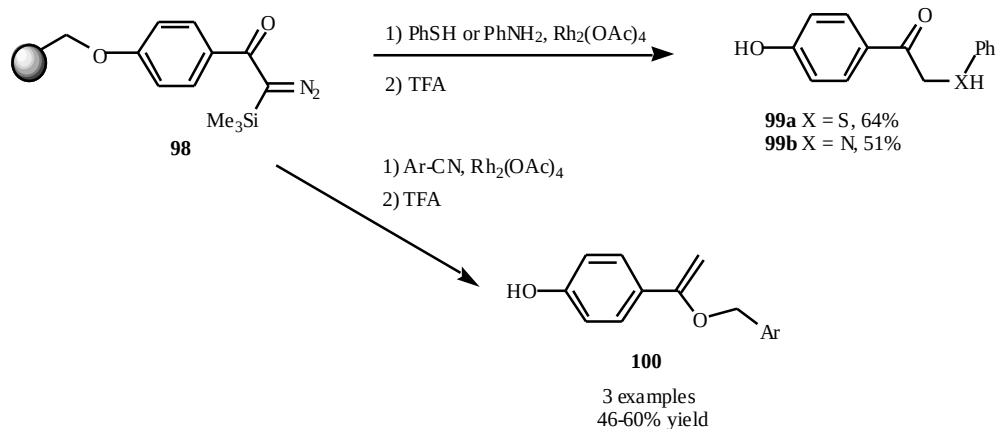
Janda and co-workers [99-102] describe the use of rhodium carbenoid N-H insertion in the formation of oxazoles, imidazolones and indoles on solid phase, using



Scheme 47. Solid phase synthesis of a branched *Leishmania* tetrasaccharide.



Scheme 48. Synthesis of stilbene-derivatives using cross-metathesis.

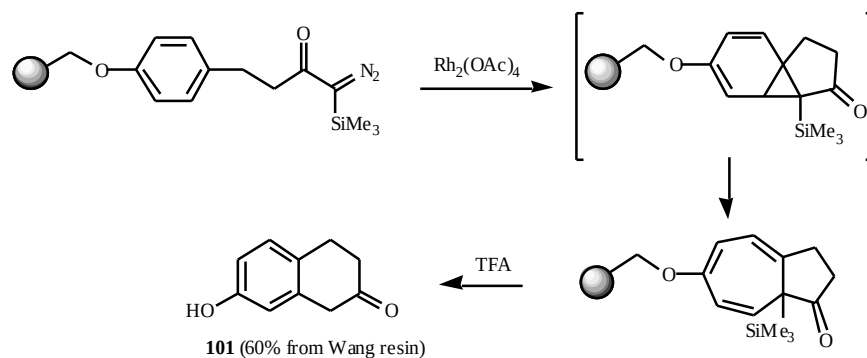


Scheme 49. Solid phase synthesis using polymer-bound diazoketones.

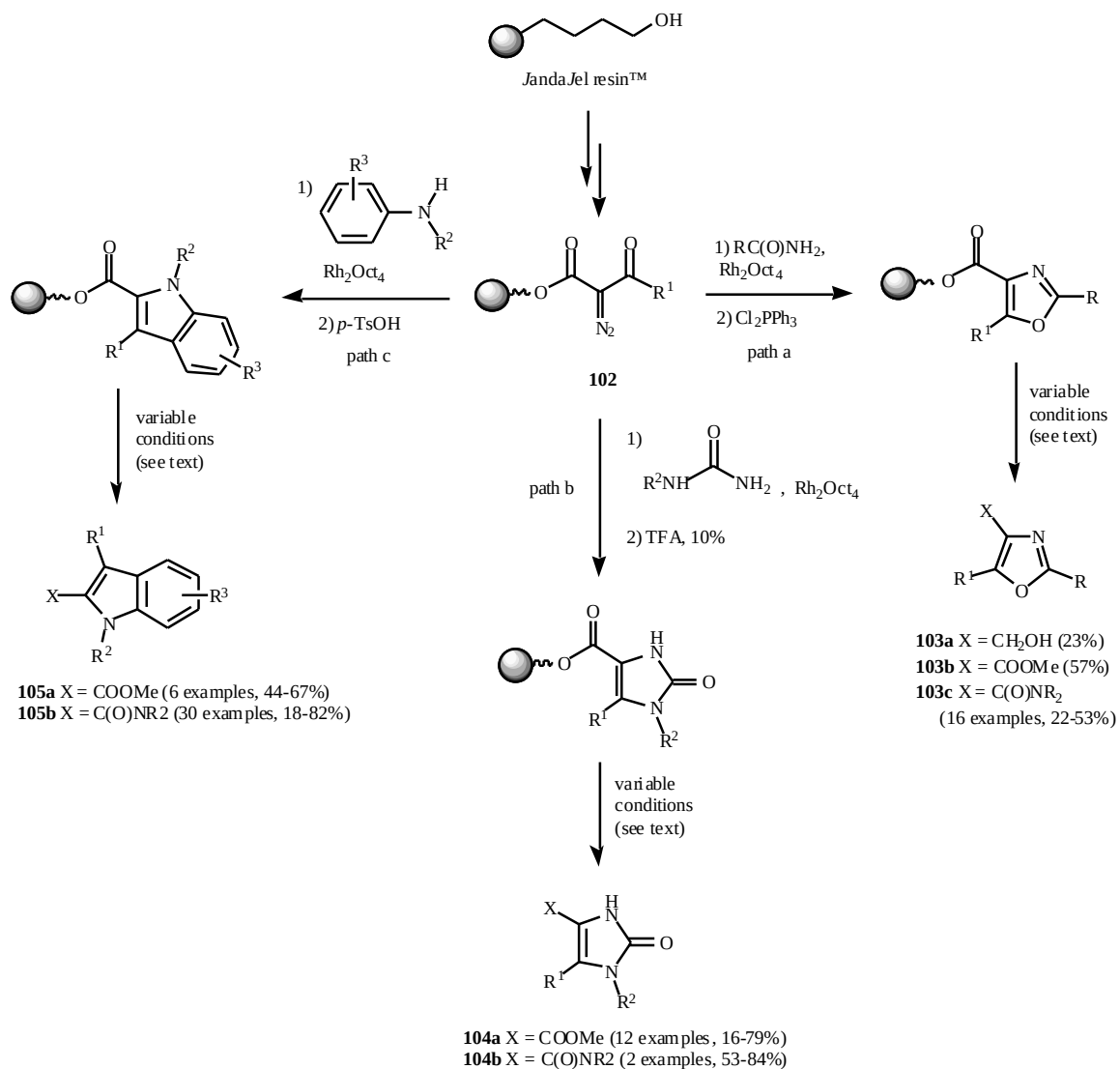
polymer-bound α -diazo- β -ketoesters (**102**) as precursors. When **102** was treated with primary amides and Rh_2OAc_4 , followed by workup with $\text{LiBH}_4/\text{LiEt}_3\text{BH}$, the hydroxy-substituted oxazole **103a** was formed (Scheme 51, path a). Cleavage with NaOMe in methanol instead afforded methyl ester **103b**, while aminolysis with a primary or secondary amine and AlCl_3 gave amides **103c** (16 examples) in high purity. If **102** was reacted with an urea derivative and Rh_2OAc_4 instead, imidazolones were formed (Scheme 51, path b). Cleavage was effected with NaOMe in methanol or aminolysis as described earlier, affording **104a** and **104b**. Reaction with substituted anilines and Rh_2OAc_4 , followed by

treatment with anhydrous *p*-toluenesulfonic acid instead gave polymer-bound indole derivatives **105**. Best results were obtained if **102** and the aniline were heated to 100°C , before the addition of the rhodium catalyst together with a second portion of the aniline derivative in toluene. Microwave heating (temperature not indicated) was also examined for the cyclisation step, but the purity of the products thus obtained was lower as compared to when conventional heating was used. Cleavage was effected as earlier yielding ester- or amide-substituted indoles.

Yamazaki and Kondo [103, 104] have applied the rhodium catalysed N-H insertion reaction in several different



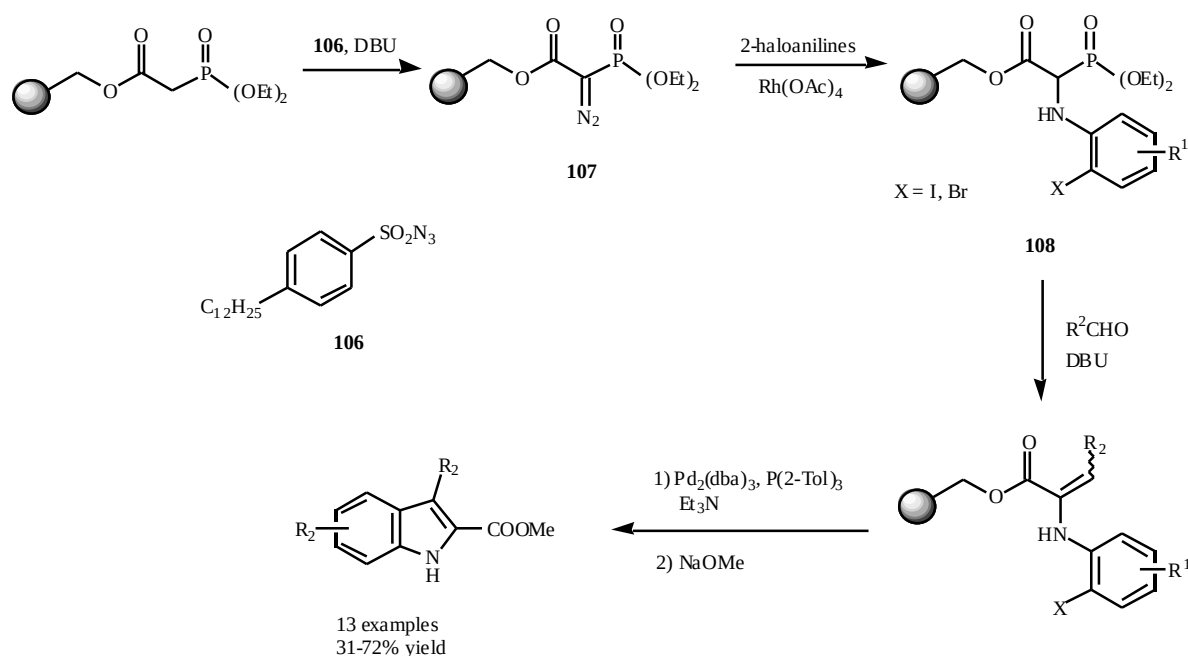
Scheme 50. An intramolecular Buchner reaction.



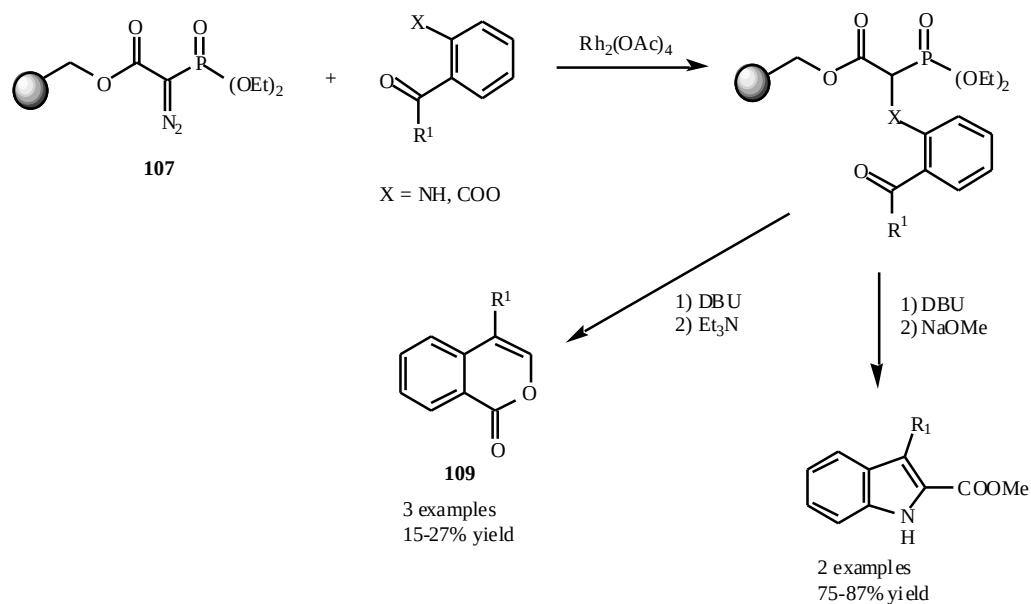
Scheme 51. Synthesis of heterocycles via rhodium carbene N-H insertion.

approaches towards the solid phase synthesis of indoles. A polymer-bound diethylphosphonoacetate (**107**, Scheme 52) was converted to the corresponding α -diazo compound using dodecylbenzenesulfonyl azide. Rhodium-catalyzed insertion of substituted iodo-anilines afforded **108** which could be transformed into an alkene precursor *via* a Horner-

Wadsworth-Emmons reaction. Addition of 20 mol% phenol was found to be beneficial for the insertion step, reducing the reaction time from 40 h to 12 h. A palladium-catalysed Heck-type cyclisation completed the sequence, yielding 2-carboxymethylindoles substituted both on the aromatic ring as well as in the 3-position.



Scheme 52. Solid phase synthesis of indoles.



Scheme 53. Synthesis of indoles and isocoumarines on solid support.

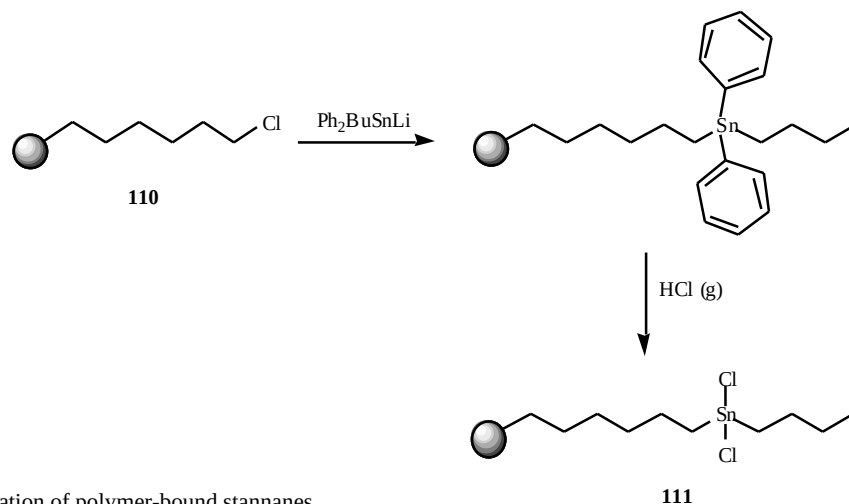
Using keto-substituted haloanilines, the indoles could be accessed directly *via* cyclisation at the Horner-Wadsworth-Emmons step, alleviating the need for the Heck-type cyclisation (Scheme 53). Isocoumarins (**109**) could also be prepared using similar methodology, starting with *ortho*-keto-substituted benzoic acid derivatives instead, although the yields were rather low in this case. Cyclisation approaches towards indoles involving palladium-catalysed amination are also described.

7. RADICAL MEDIATED REACTIONS

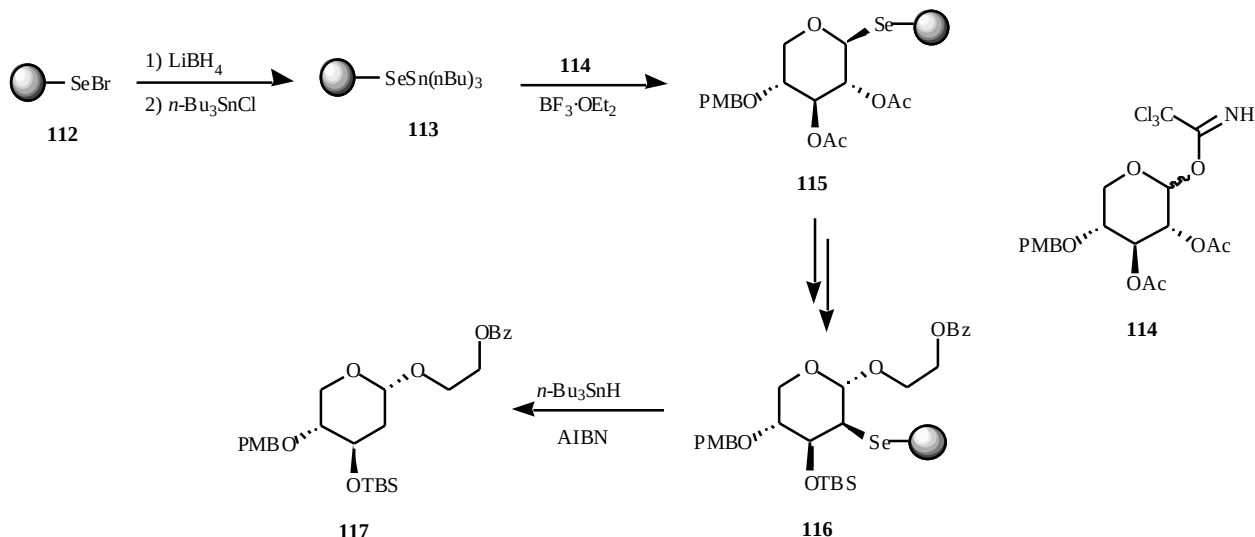
Martins and co-workers [105] prepared a polymer-bound butyldichlorostannane according to Scheme 54. A

polystyrene resin (Amberlite XE305) was lithiated and reacted with the appropriate dihalide to give the desired chlorobutylpolystyrene **110**. The polymer-bound alkyl chloride was stannylated with Ph₂BuSnLi, and the phenyl groups subsequently replaced by chloride to form the polymer-bound dichlorobutylstannane **111**. Reaction progress was followed by 1D HR-MAS ¹H NMR and ¹¹⁹Sn NMR under MAS conditions. ¹¹⁹Sn NMR is a powerful tool for monitoring reactions on polymer-bound diamagnetic Sn-species, clearly demonstrated in the report by Martins et al.

The use of selenium-based linkers for solid phase synthesis offers a traceless approach where the linker can be cleaved *via* a radical mechanism using *n*-Bu₃SnH and a



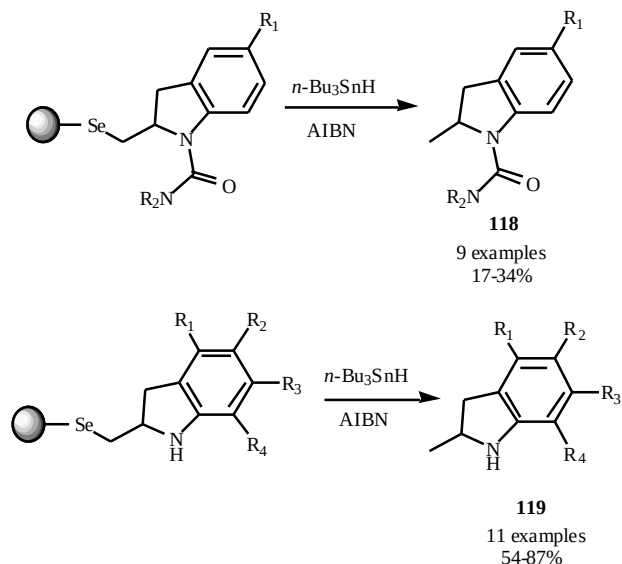
Scheme 54. Preparation of polymer-bound stannanes.



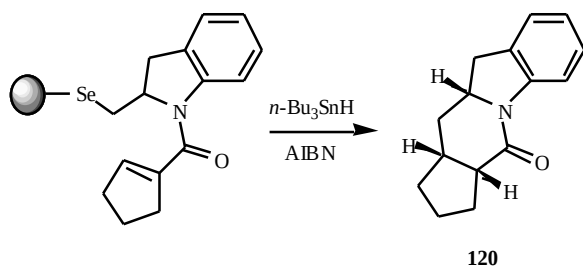
Scheme 55. Synthetic studies towards Everninomycin and derivatives.

radical initiator. Scheme 55 shows parts of the combinatorial effort towards the synthesis of Everninomycin, reported by the Nicolaou group [106], where only the organometallic steps are highlighted for simplicity. Seleniumbromide polystyrene **112** was subjected to reductive conditions using LiBH_4 , affording the corresponding lithioselenide resin, which was quenched with an excess of $n\text{-Bu}_3\text{SnCl}$ yielding seleno-tin resin **113** (should be used freshly prepared). Addition of trichloroacetimidate **114** under Lewis-acidic conditions completed the sequence, forming **115** in 90% yield via cleavage with $n\text{-Bu}_3\text{SnH}$ /AIBN. After four more steps on solid phase, the selenium-carbon bond of **116** was cleaved reductively with $n\text{-Bu}_3\text{SnH}$ /AIBN in benzene at elevated temperature giving **117** in excellent yield. Oxidative processes involving the polymer-bound intermediate **116** were also studied.

Another example by the same group [107], also involving a selenium based linker, can be found in the synthesis of a indoline library, where the compounds cleaved with $n\text{-Bu}_3\text{SnH}$ /AIBN in toluene at elevated temperature give indolines in good to excellent purities after extractive workup (Scheme 56). The *N*-acylated products (**118**) were



Scheme 56. Solid phase synthesis of indolines using a selenium linker.



Scheme 57. A radical-mediated cyclisation-cleavage strategy to form polycyclic indolines.

formed in lower yields and purities compared to the free NH indolines (**119**), probably due to the extra steps needed for the generation of the *N*-substituted library.

The indoline nitrogen was also used to link an olefinic radical acceptor to give the tetracyclic indoline **120** as a single diastereomer in 26% overall yield (Scheme 57). Three more examples of the same type were also successful, giving radical-cyclised products.

A nitrile derivative of the solid phase-bound indoline was treated with an excess of Me_3SnN_3 in toluene at 100 °C for 12h to afford the free tetrazole, which was alkylated and then cleaved with hydrogen peroxide instead of $n\text{-Bu}_3\text{SnH/AIBN}$, giving twelve indoline-tetrazole derivatives (**121**) in low yields but high purities (Scheme 58).

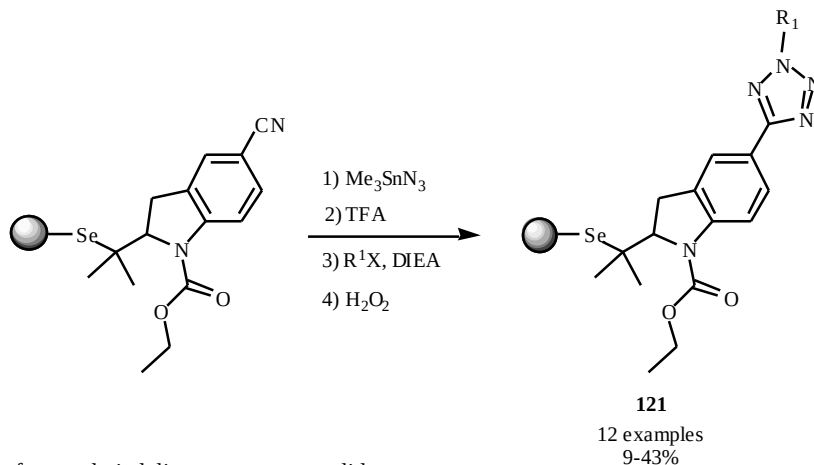
The polymer-bound *b*-acylsulfone **122** in Scheme 59 was reductively cleaved with an excess of SmI_2 to yield the corresponding olefin as an *E/Z* mixture, as reported by D'herde and De Clercq [108]. The best selectivity was found

for $\text{R}^1 = \text{CH}_2\text{C}_6\text{H}_{11}$ and $\text{R} = \text{Bz}$, when using DMPU as an additive. Under these conditions, the selectivity was 94:6 in favor of the *E*-isomer.

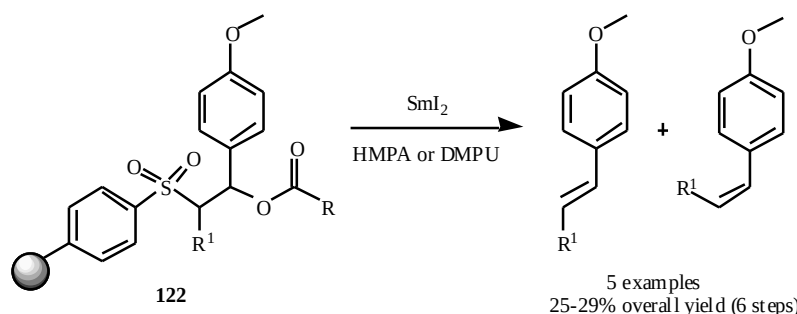
Bräse and coworkers [109] reported a solid phase synthesis of aryl ethers using the Nicolaou variant of the Ullmann reaction. Polymer-bound aryl bromides **123**, anchored to the solid phase via a triazene linkage, were subjected to modified Ullman conditions as shown in Scheme 60, using an excess of the phenolic component. No reaction occurred when using electron-poor phenols, indicating the importance of the nucleophilicity in this reaction on solid phase. The compounds were liberated from the polymer using TFA/ TMSN_3 in DCM. A 32-member library was prepared using this methodology with yields ranging from 34% to 95%.

The use of SmI_2 in solid phase synthesis is being extended in accordance with its utility in solution phase reactions. SmI_2 has been shown to reductively cleave N-O, C-O, C-N and C-S in solution phase, indicating the possibilities of using SmI_2 to effect traceless cleavage. One example of this strategy is shown in Scheme 61, where Abell and co-workers [110] used a Wang-type hydroxylamine-linker, which is normally cleaved with TFA to generate the hydroxylamide, but in this case instead liberating the free amide **124** upon reductive cleavage with SmI_2 in THF. 8 different amides were cleaved reductively from the solid support, with yields ranging from 31% to 54% over three steps (acylation, alkylation and cleavage) and purities in most cases between 84% to 99% as determined by HPLC.

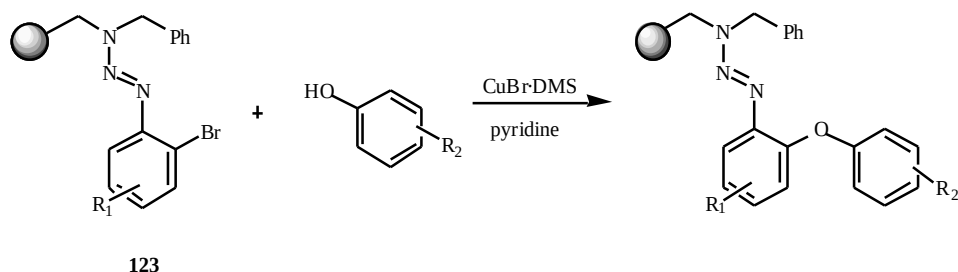
In the solid phase synthesis of a 1,4-diazepan-2,5-dione, a chlorotriptyl-linked hydroxylamine was used successfully



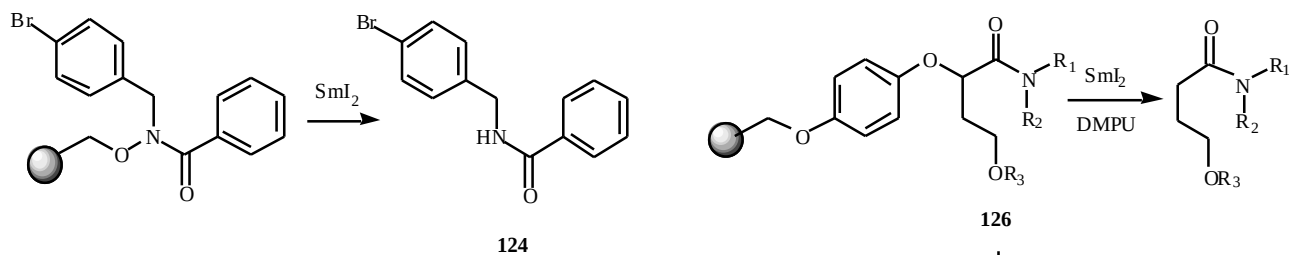
Scheme 58. Synthesis of tetrazole-indoline structures on solid support.



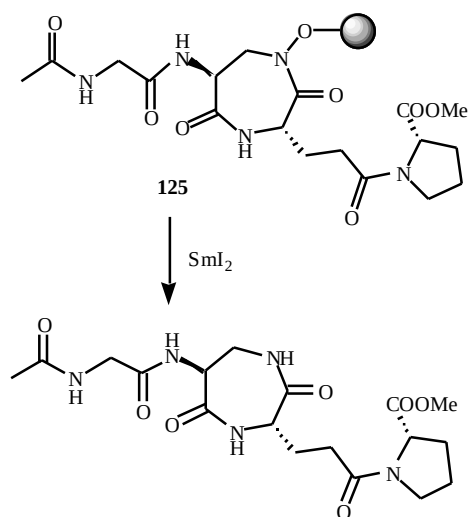
Scheme 59. Julia-Lythgoe olefination on solid phase.



Scheme 60. A copper-catalysed Ullman coupling on solid phase using a triazene linker.



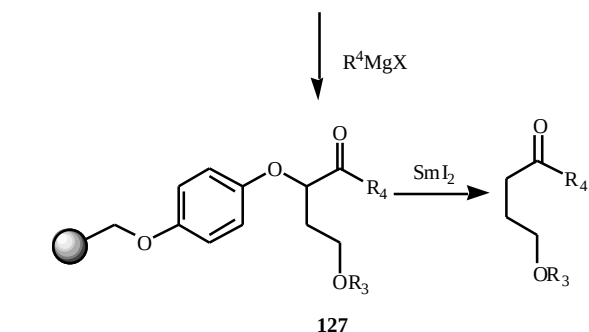
Scheme 61. Reductive traceless cleavage of a hydroxylamine linker with samarium iodide.



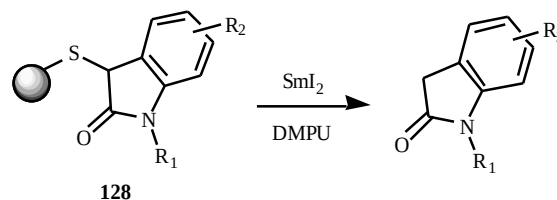
Scheme 62. Samarium-mediated cleavage of a peptidomimetic structure.

by Taddei and co-workers [111] (Scheme 62). Upon cleavage of the polymer-bound substrate **125** with SmI_2 , the corresponding free amide was formed in 51% yield. A TFA cleavage of **125** instead gave 75% of the corresponding hydroxylamide. With the SmI_2 -induced cleavage, an aqueous workup and a short path silica gel column was needed which may explain the lower yield.

In the solid phase reduction of α -aryloxycarbonyls, Procter and co-workers [28] developed a traceless linker as outlined in Scheme 63. The linker used in the solid phase sequence was a Bromo Wang resin, elongated with a phenol. The cleavage of the amide derivatives **126** was more difficult than the corresponding cleavage of the ketones **127**. The use of DMPU as an additive was found to be beneficial to the reaction. Five amides and three ketones were synthesised using this method. The cleavage conditions tolerated ester functionality as well as TBDMS ethers, indicating the nature



Scheme 63. Cleavage of α -aryloxycarbonyl compounds.

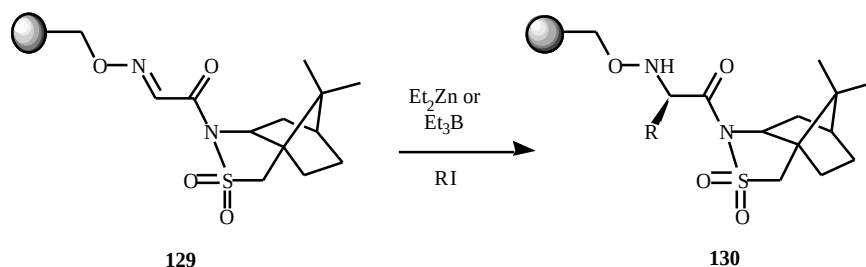


Scheme 64. Solid phase synthesis of oxindoles.

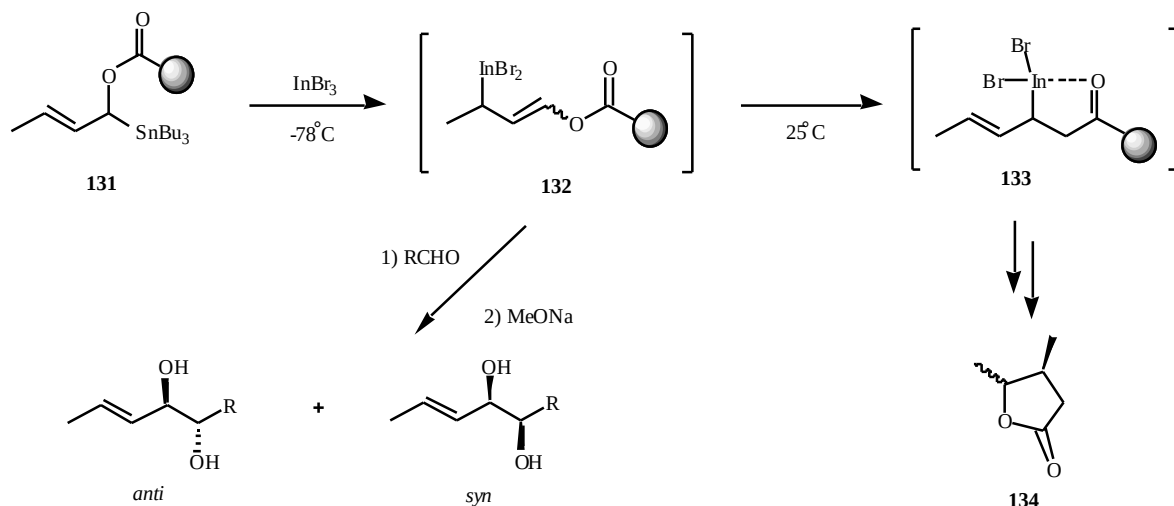
of the cleavage to be mild, neutral and chemoselective. The yields ranged from 16% to 31% with purities in the range of 70% to 93%.

A similar approach was used by the same group [112] for the construction of oxindoles (Scheme 64). An α -bromo arylamide was reacted with a benzyl-thiol resin derived from Merrifield resin, whereupon the polymer-bound substrate was activated for the Pummerer cyclization, giving the desired polymer-bound oxindole **128**. The cleavage was subsequently conducted using the same protocol as for the aryloxy amides described previously. The oxindoles were synthesised in high purities and moderate yields, and in contrast to the earlier report, no aqueous workup was needed.

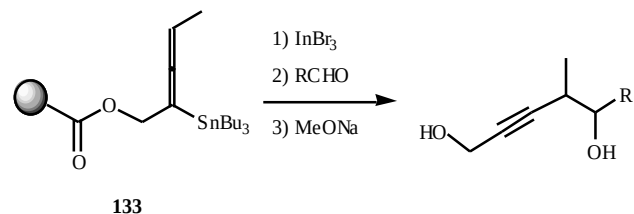
The use of Et_2Zn as a radical initiator has been employed successfully on solid phase by Naito *et al.* [113], using a polymer-bound version of Oppolzer's camphorsultam derivative (Scheme 65). The use of this chiral auxiliary enables stereocontrol of the alkyl radical addition. Oxime



Scheme 65. Stereoselective radical addition to a polymer-bound oxime ether.



Scheme 66. Organoindium chemistry on solid phase.



Scheme 67. Allenyldindium reagents in the synthesis of homopropargylic alcohols.

ether **129** was treated with three different alkyl halides (Et, *i*-Pr, Cy) and with Et_2Zn or Et_3B as the radical initiator. Cleavage from the polymeric support with TFA in DCM gave **130** in good yields (41–74%) and excellent diastereoselectivity (de 90–95%). The yield was slightly higher for all three entries when using Et_3B as the initiator. For the bulkier halides (*i*-Pr, Cy) Et_2Zn or Et_3B compete in the addition reaction, favoring the ethylated product. However, by using an excess of reagent, chemoselectivity was obtained.

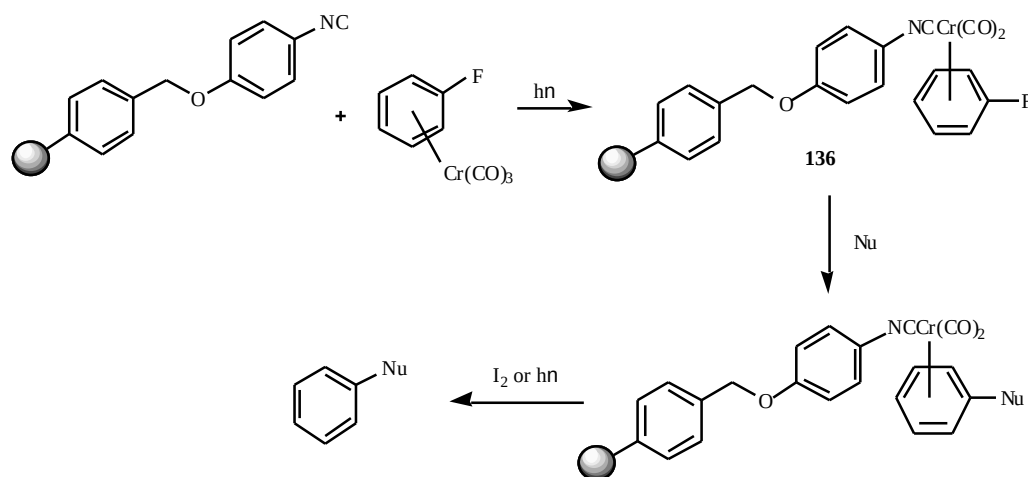
In the solid phase synthesis of 1,2-diols and gamma-lactones, Cossy *et al.* [114] used an allylic addition of aldehydes to an in situ generated organoindium species (Scheme 66). The polymer-bound stannane **131** was synthesised from carboxypolystyrene and the corresponding hydroxyl stannane *via* a standard esterification reaction. **131** was then treated with InBr_3 at -78°C and subsequently

reacted with 5 different aldehydes to give the *syn* and *anti* diols in good to excellent yields. Almost equal amounts of the *syn/anti* products were found with reaction of aromatic or a conjugated aldehyde, while the *anti* product was formed exclusively when aliphatic aldehydes were used. However, when the organoindium complex **132** was allowed to reach 25°C , the more thermodynamically stable complex **133** was formed and upon reaction with aldehydes and subsequent cleavage with MeONa, the corresponding lactols **134** were formed in good yields.

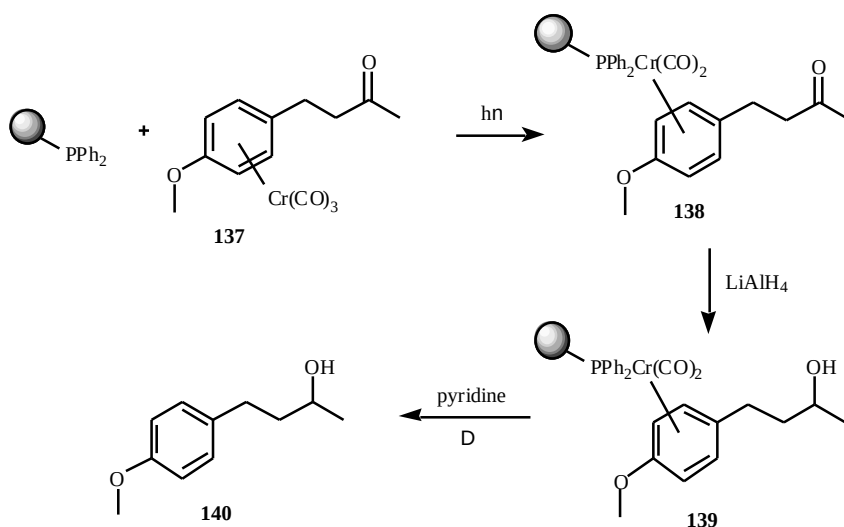
Another report by the Cossy group [115] is highlighted in Scheme 67, where a polymer-bound allenyltin reagent is transmetallated to give the allenylindium reagent. The polymer-bound allenyltin reagent **135** was prepared *via* attachment of the corresponding alcohol to carboxypolystyrene. **135** was then transmetallated with InBr_3 , reacted with aldehydes and finally cleaved with MeONa to give the homopropargylic alcohols in moderate to high yields. The polymer-bound allenyltin can react directly with aldehydes in the presence of a Lewis acid to give the same products.

7. TRANSFORMATIONS INVOLVING METAL CARBONYL REAGENTS

Maiorana, de Magistris and co-workers [116] evaluated the polymer-bound fluorobenzene chromium dicarbonyl isocyanide **136** in nucleophilic aromatic substitution reactions (Scheme 68). The substrate was linked to the solid phase *via* an isonitrile ligand, thus activating the aromatic



Scheme 68. Nucleophilic aromatic substitution on solid phase using chromium arene complexes.



Scheme 69. A phosphine-chromium linker for arenes.

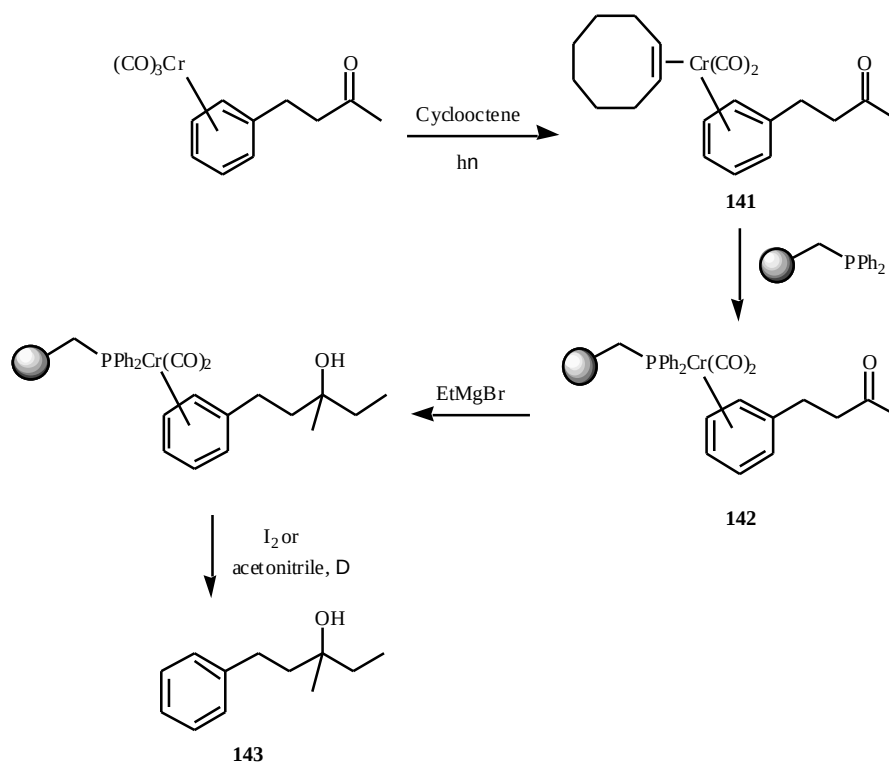
ring towards nucleophilic attack; use of a phosphine linkage would instead deactivate the aromatic fluoride position according to similar studies in solution. The fluorobenzene chromium tricarbonyl complex was attached to the solid support *via* direct photochemical irradiation, whereupon **136** was subjected to treatment with three different amines. According to ^{19}F NMR, full conversion was achieved. The products were cleaved from the resin using iodide in CH_2Cl_2 for 1 h or exposed to sunlight and air for 2 days, giving good overall yields. The scope of the reaction was examined further in a following study [117], where the substitution pattern of the polymer-bound arene chromium dicarbonyl isocyanide was varied.

Gibson *et al.* [118] used a polymer-supported phosphine ligand to examine the use of σ -bound ligands on solid phase (Scheme 69). The use of commercially available polymer-bound phosphine makes this route appealing due to the relatively low cost of the resin. The polymer-bound phosphine was irradiated together with the arene chromium tricarbonyl complex **137** to give the polymer-bound complex **138** in 40% yield according to ^{31}P NMR. The ketone functionality was reduced with LiAlH_4 to form complex **139**

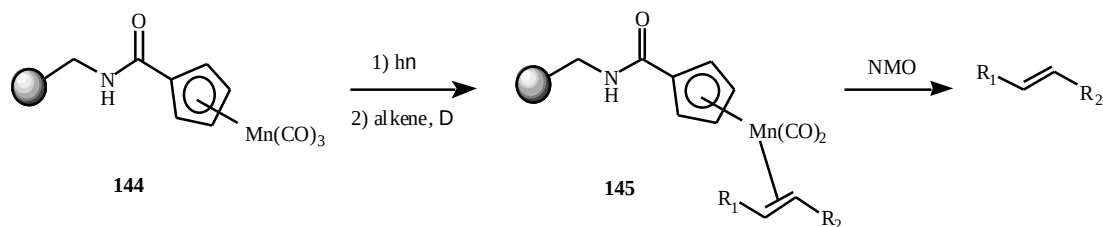
with a loading of 25%, although the ketone was completely consumed in the reaction as observed by IR. Decomplexation and liberation from the polymer was performed in pyridine at elevated temperature, yielding alcohol **140** in 90% yield. α -alanine was also immobilised successfully using similar methodology, and subjected to peptide coupling.

An alternative route to immobilise arene chromium tricarbonyls, reported by Rigby [119], is to use a stepwise protocol as shown in Scheme 70. A carbonyl ligand was replaced by cyclooctene *via* photolysis, affording complex **141**, followed by treatment with polystyrene-supported phosphine to form **142** in a high yield as determined by weight and *via* cleavage. The ketone was reacted with ethyl magnesium bromide to give the tertiary alcohol, and cleavage was effected using iodide (*via* oxidation) or acetonitrile (*via* ligand displacement) affording alcohol **143** in 79% yield. Decomplexation with I_2 gave lower yields due to the more intricate workup needed. Alcohols and esters were also successfully immobilised using the described protocol.

A manganese linker strategy for alkenes (Scheme 71) using amide-linked cymantrene (cyclopentadienyl



Scheme 70. Traceless solid phase synthesis using a chromium-phosphine linker.



Scheme 71. A cyclopentadienyl manganese tricarbonyl linker developed for alkenes.

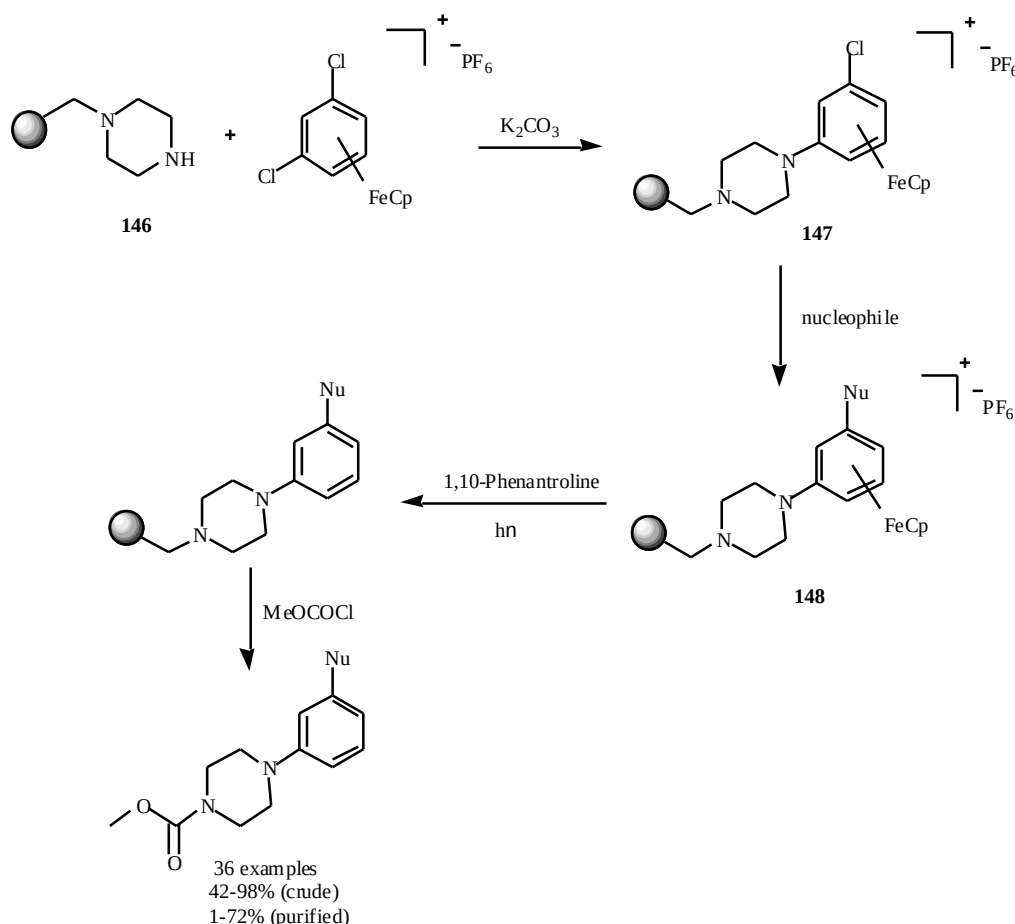
manganese tricarbonyl) resin was developed by Zhang and Lepore [120]. Manganese complex **144** was either prepared using cymantrene acid chloride and a polymer-bound amine or *via* the reaction of lithio-cymantrene and polymer-bound isocyanate. The reaction with the acid chloride gave a higher loading, but large-scale synthesis of the acid chloride was problematic. Once formed, polymer **144** was irradiated with UV-light at 254 nm in THF whereupon one of the cymantrene carbonyls was replaced by THF forming a deep red complex. The ligand exchange could be monitored by IR. The intermediate complex then readily coordinated the desired alkene forming **145**. Oxidation with NMO release the alkene in a >20% recovery and a purity of 95%.

Ruhland *et al.* [121] used a piperazine-Merrifield resin to anchor a $(\text{Fe(II)CpPh})^+\text{PF}_6^-$ complex for $\text{S}_{\text{N}}\text{Ar}$ reactions on solid phase (Scheme 72). Polymer-bound piperazine **146** was reacted with a dichloro[(cyclopentadienyl)-benzene] $\text{Fe(II)}^+\text{PF}_6^-$ complex in the presence of potassium carbonate in THF at elevated temperature, forming **147**. The residual chloride functionality was subsequently reacted with different nucleophiles (oxygen, sulphur, selenium, nitrogen and phosphorus) using different protocols giving the substituted

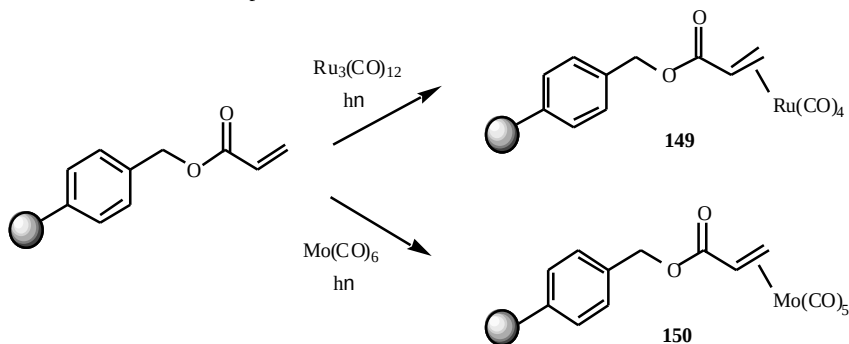
polymer-bound complex **148**. The decomplexation was carried out using a ligand-ligand exchange reaction developed by Pearson. In the final step, the alkyl piperazine was cleaved from the resin using methyl chloroformate in dichloroethane to give aryl piperazine carbamates in good yields.

Leadbeater and Sharp [122] immobilised ruthenium and molybdenum carbonyls on polystyrene and used in a ligand-exchange protocol. A REM-linker was reacted with $\text{Ru}_3(\text{CO})_{12}$ and $\text{Mo}(\text{CO})_6$ respectively to give polymer-bound organometallic complexes **149** and **150** (Scheme 73). The polymer-bound ruthenium complex was prepared in 20% yield according to ICP-AA and mass increase, while the corresponding molybdenum complex was formed in 77% yield. The use of a polymer-bound alkene ligand increases the stability of the complex. Cleavage was effected *via* a ligand-exchange reaction with PPh_3 in CH_2Cl_2 at elevated temperature forming $\text{RuPPh}_3(\text{CO})_4$ in 75% and $\text{MoPPh}_3(\text{CO})_5$ in 85% yield.

Another use of a ligand-exchange reaction on solid phase is the $\text{Mo}(\text{CO})_6$ mediated palladium-catalysed carbonylation reported by Yamazaki and Kondo [123] and highlighted in



Scheme 72. Iron-assisted S_NAr reactions on solid phase.



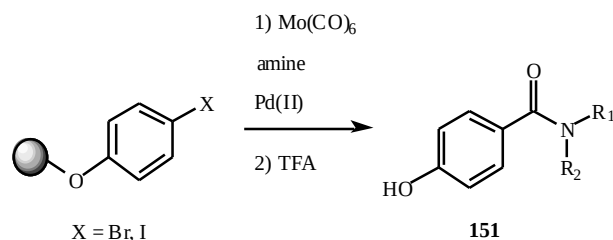
Scheme 73. Preparation of polymer-bound ruthenium and molybdenum carbonyl complexes.

Scheme 74. A *para*-halophenol attached to Wang-resin was carbonylated with $\text{Mo}(\text{CO})_6$ and two different amines (benzylamine and morpholine). The aryl amide **151** was liberated from the polymer using TFA in DCM to give the aryl amides in excellent yields and purities. Three more examples are reported.

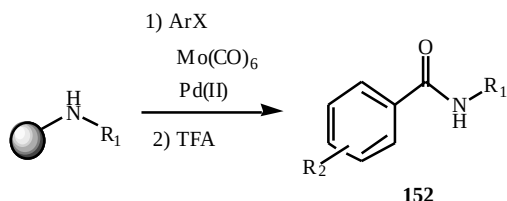
The reverse reaction in terms of aryl halide/amine was also successful (Scheme 75). Five different linkers (Rink, PAL, Sieber, FMPE and Indolyl) were used in conjunction with 4-iodoanisole to produce amides **152** in moderate to good yields and excellent purities, apart from the Sieber linker which gave poor results overall. A similar protocol was used to prepare isoquinolone **153** (Scheme 76).

In another report by the Gibson group [124], a polymer-bound cobalt carbonyl complex was used as a linker for alkynes (Scheme 77). Complex **154** was reacted with polymer-bound diphenylphosphine (1.6 mmol/g) to give a mixture of the mono- and diphosphine-cobalt complexes **155a** and **155b**. The 3:2 ratio, as well as the loading (0.94 mmol/g), was determined by ^{31}P NMR and IR in favor of the monophosphine complex. Complexes **155** were subsequently applied in an oxidation reaction, forming **156**, and an acylation reaction to demonstrate their applicability.

A Pauson-Khand annelation was also performed using the same linking method (Scheme 78). The cobalt complex **157** was reacted with diphenylphosphine polystyrene to give



Scheme 74. Carbonylation of polymer-supported aryl halides with molybdenum carbonyl.



Scheme 75. Alternative approach to amides via carbonylation.

the immobilised mono- and bisphosphine complexes **158a** and **158b** in a ratio of 3:2 in favor of the monophosphine. The Pauson-Khand reaction was performed under 1.05 bar CO at elevated temperature to yield the annelated product **159** in 26% yield after chromatography.

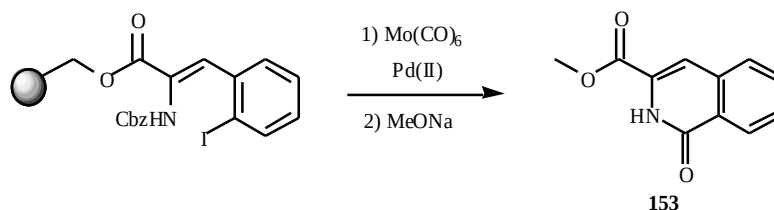
Schreiber *et al.* [125] used a Pauson-Khand strategy to generate rigid and highly functionalised tricyclic compounds in high purities (Scheme 79). The major part of the 2500 library displayed a purity above 80% according to LC-MS

analysis. The Pauson-Khand reaction was performed in a stepwise manner, involving the complexation of **160** with $\text{Co}_2(\text{CO})_8$ in CH_2Cl_2 followed by treatment with NMO to afford cyclopentenones **161**. Conjugate addition and subsequent cleavage with HF completed the sequence.

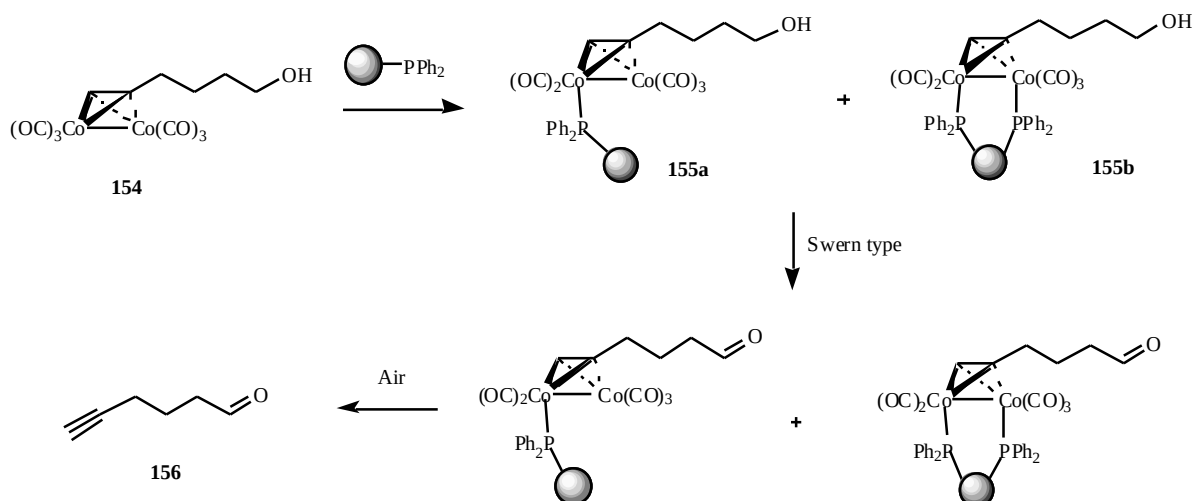
Cassel *et al.* [126] synthesised alkynyl-derivatives using a Nicholas reaction on solid phase as the key step (Scheme 80). Polymer-bound alkynes **162** were reacted with $\text{Co}_2(\text{CO})_8$ in CH_2Cl_2 to give the solid-supported cobalt-alkyne complexes **163**, which were subsequently treated with carbon or oxygen nucleophiles in the presence of a Lewis acid in a Nicholas reaction [127] to afford the derivatised alkynes **164** in good overall yields (4-5 steps). Both carbon and oxygen nucleophiles performed equally well. However, no product was formed when $\text{R}^1 = \text{R}^2 = \text{Me}$, probably due to a competing elimination pathway.

A carbon monoxide insertion developed by Cordero and Brandi [128] is shown in Scheme 81 where the polymer-bound isoxazolidine **165** was treated with $\text{Mo}(\text{CO})_6$ to give α -hydroxyamide **165**.

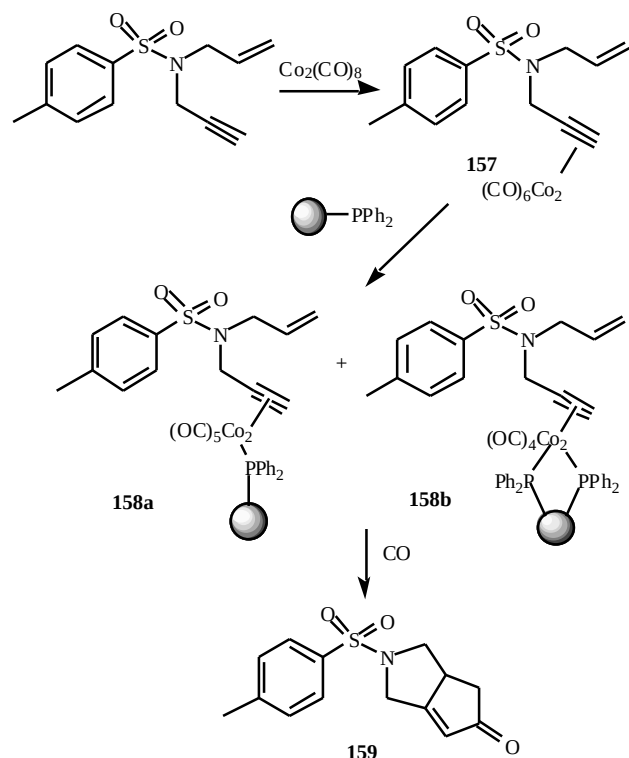
The hydroformylation reaction [129], i.e. the conversion of alkenes to aldehydes using carbon monoxide, hydrogen and a transition metal catalyst, is one of the most important organic reactions applied in the chemical industry. One problem with the oxo-process has been the difficulty in separating the product aldehyde from the organometallic reagent used, and one way of solving this problem could be to attach the substrate alkene to a solid support instead. Marchetti *et al.* [130-132] have studied both these methods, but only the latter approach will be discussed here. Different



Scheme 76. Isoquinolone synthesis on solid support.



Scheme 77. A polymer-bound cobalt carbonyl complex as a traceless linker for alkynes.



Scheme 78. A Pauson-Khand reaction on solid support using a cobalt-tethered alkyne.

alkenes were attached to pre-activated chlorotrityl resin and subjected to hydroformylation conditions in the presence of a

rhodium catalyst (Scheme 82). For linear alkenes **166**, using $\text{Rh}(\text{CO})_2(\text{acac})$ as the source of rhodium, the conversion to the desired polymer-bound aldehyde was generally >99%, with a linear:branched ratio pending between 23:77 and 52:48. The use of $\text{HRh}(\text{CO})\text{PPh}_3$ in conjunction with xantphos favoured formation of the linear aldehyde.

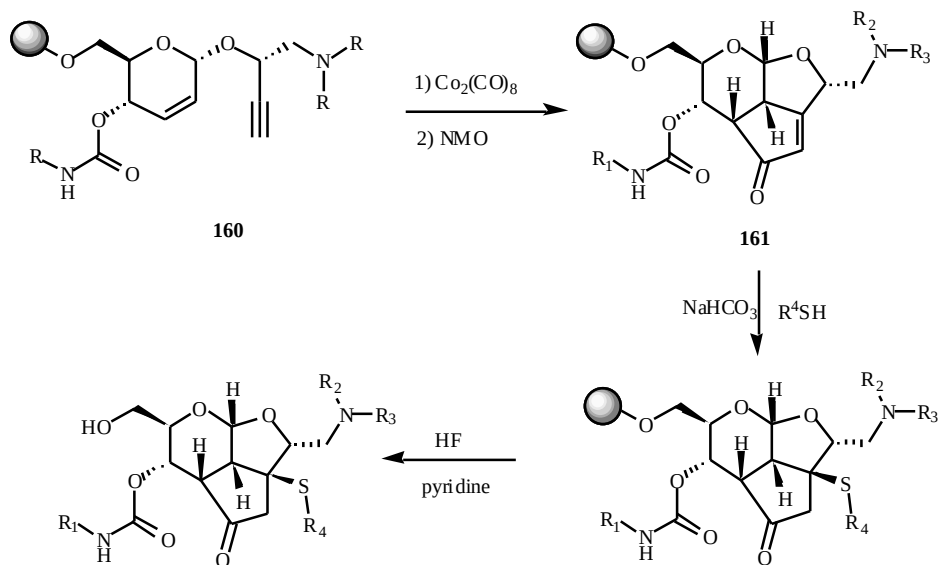
Marchetti also exploited the combined hydroformylation/reductive amination sequence in a one-pot domino reaction to form the corresponding **167** amines in good yields (an example is shown in Scheme 83).

8. MISCELLANEOUS REACTIONS

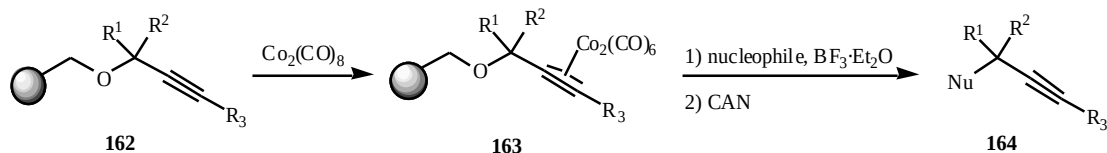
Sun *et al.* [133] reported a rhodium-catalysed [2+2+2] cycloaddition in the solid phase synthesis of isoindolines **168** (Scheme 84). Attaching the substrate to a solid phase avoided competing linear dimerisation and cyclo-trimerisation which usually accompanies these types of cyclisations in solution.

Wilkinson's catalyst has also been used by Takahashi [134] in the reduction of enones attached to ArgoGel™ in the synthesis of phlorizin derivatives **169** (Scheme 85).

Takahashi [135] also used chiral rhodium catalysts in the asymmetric hydrogenation of unsaturated amides for the synthesis of dipeptides containing unnatural amino acids (Scheme 86). A library of dipeptides was prepared on Rink amino Crowns, where glycine, alanine or leucine was attached to the resin, followed by a sequence of Horner-Wadsworth-Emmons and Mizoroki-Heck reactions, affording a derivatised dehydroalanine moiety (**170**). Asymmetric hydrogenation using $[\text{Rh}(\text{COD})\text{L}^+]\text{ClO}_4^-$, with (*R,R*)- or (*S,S*)-Me-DuPhos as the ligand L, and subsequent

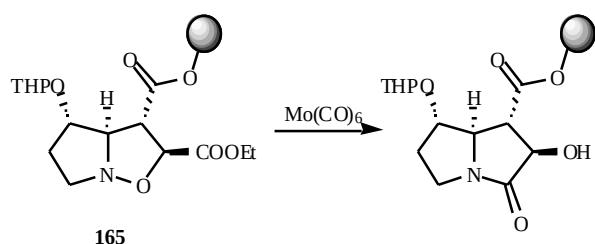


Scheme 79. Diversity-oriented synthesis using a Pauson-Khand reaction.

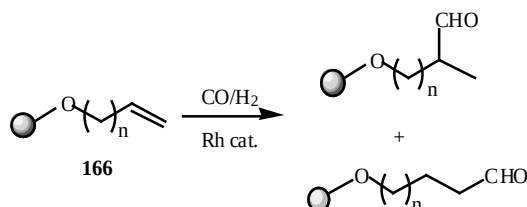


Scheme 80. A solid phase Nicholas reaction yielding derivatised alkynes.

12 examples
25-56% (4-5 steps)



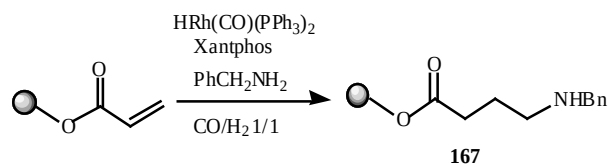
Scheme 81. Carbonylation of isoxazolidines using molybdenum carbonyl.



Scheme 82. Hydroformylation on solid phase.

cleavage furnished a library of 36 dipeptides of high diastereomeric purity.

Liao *et al.* [136] have performed homocoupling of resin-bound acetylenes (**171**) catalysed by a combined AgOTs-CuCl₂-TMEDA system, where Ag(I) activates the triple bond *via* π -complexation, thus facilitating the Cu(II)-mediated oxidative coupling (Scheme **87**). Three different



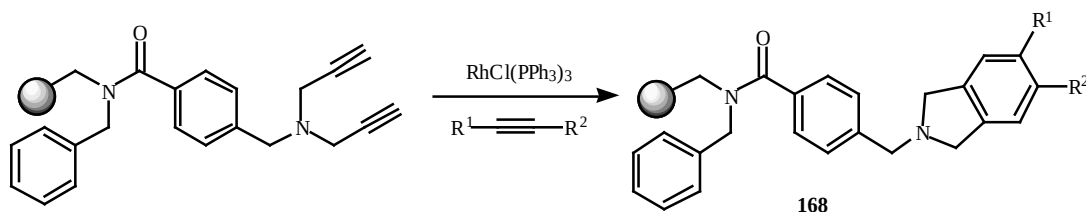
Scheme 83. Hydroformylation and reductive amination on solid phase.

silver salts were tested, with AgOTs giving the best results.

Austin and co-workers [137, 138] have earlier reported a solid phase 1,3-dipolar cycloaddition of isomünchnones with vinyl ethers. The same group [139] recently report an extension of this work, involving the use of a chiral linker based on α -hydroxyvaline (Scheme 88). Reaction of **172** with a rhodium catalyst and vinyl ethers, followed by cleavage *via* aminolysis, furnished bicyclic compounds **173** in high enantiomeric excess. Both enantiomers of the compounds could be prepared by starting with (*R*)- or (*S*)- α -hydroxyvaline. The cleavage was chemoselective in that only the cyclised product was cleaved from the resin, indicating involvement of the bicyclic intermediate in the cleavage step.

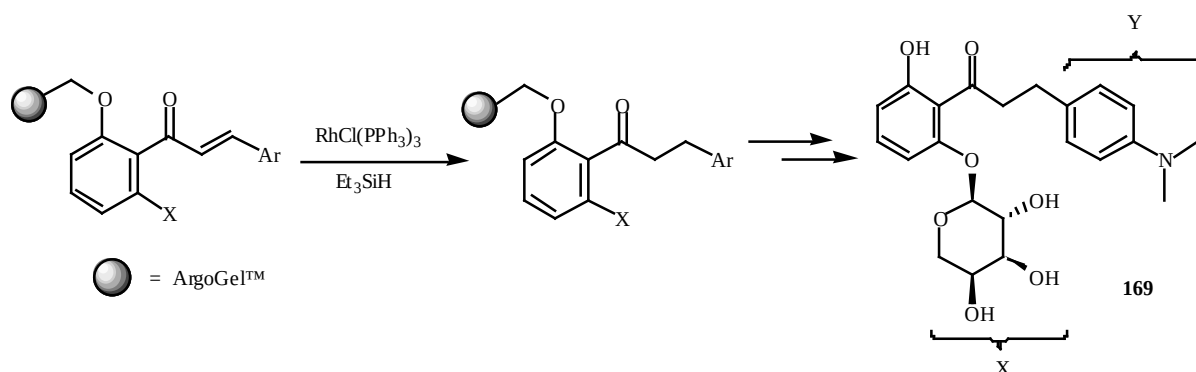
CONCLUSIONS AND OUTLOOK

The use of solid phase organometallic methodology is in many cases advantageous over the corresponding solution phase counterparts in terms of avoiding dimerisation and



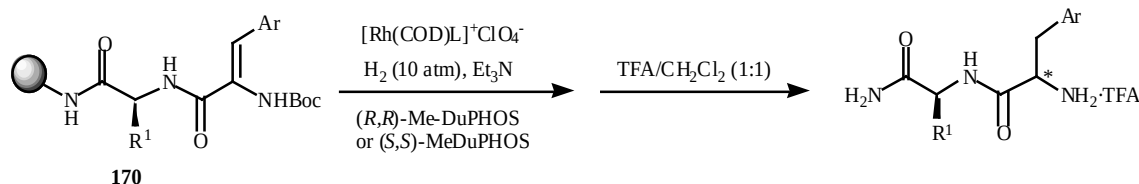
Scheme 84. A rhodium-catalyzed [2+2+2]-cyclisation on solid phase.

8 examples (20-85%)

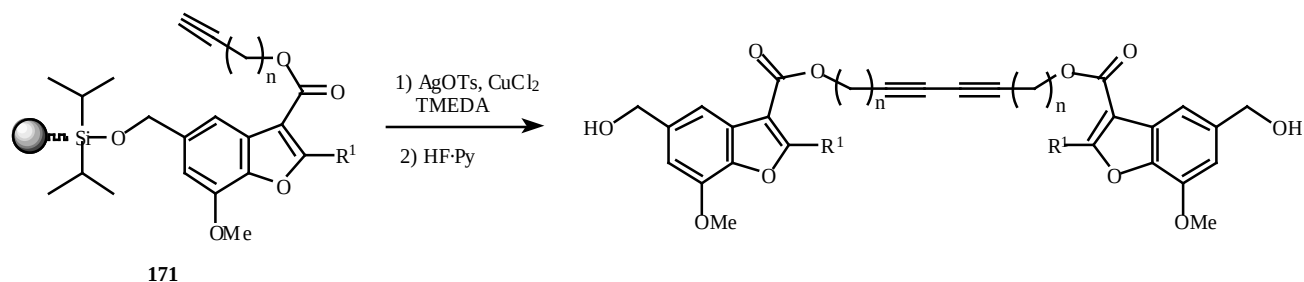


Scheme 85. A phlorozine library prepared by hydrosilylation.

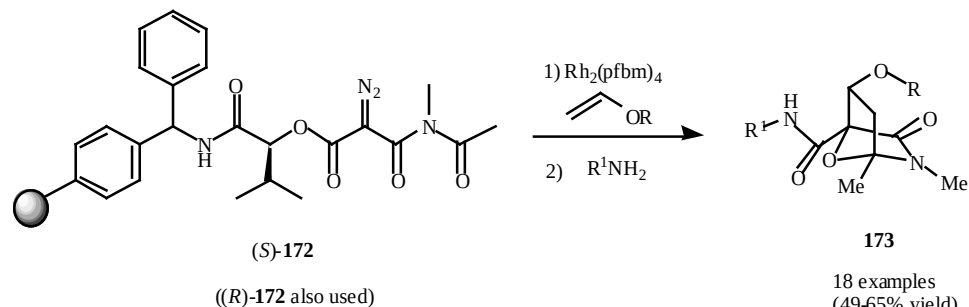
132 compounds with variations in X and Y



Scheme 86. Asymmetric hydrogenation on solid phase.



Scheme 87. Oxidative coupling of polymer-bound acetylenes.



Scheme 88. Rhodium-catalysed synthesis of bicyclic compounds.

homocoupling that often accompany reactions in solution. Purification of the final product is also simplified, and the contamination by metal impurities *via* leaching, that in some cases accompanies the use of polymer-bound organometallic reagents, is avoided. Recent developments have seen an expansion of available methodologies for solid phase organometallic transformations, and many of these will be of great use to organic as well as medicinal chemists. Nevertheless, there are areas where much work still remains to be done, for instance in the field of asymmetric organometallic synthesis on solid phase.

ACKNOWLEDGEMENTS

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ABBREVIATIONS

acac	=	acetylacetonate
AIBN	=	2,2'-azo-bis-isobutyronitrile
Boc	=	<i>tert</i> -butoxycarbonyl
BtH	=	1H-benzotriazole
CSIm ₂	=	1,1'-thiocarbonyl-diimidazole
Cy	=	cyclohexyl
dba	=	dibenzylideneacetone
DBU	=	1,8-diazabicyclo[5.4.0]undec-7-ene
DIBAL-H	=	diisobutyl aluminumhydride
DIEA	=	diisopropylethylamine
DMM	=	dimethylmaleoyl
DMPU	=	1,3-dimethyl-3,4,5,6-tetrahydro-2(1H)-pyrimidone
DMS	=	dimethyl sulfide

dppf	=	1,1'-bis(diphenylphosphino)ferrocene
Fmoc	=	9-fluorenyl methoxycarbonyl
HBTU	=	O-benzotriazol-1-yl- <i>N,N,N',N'</i> -tetramethyluronium hexafluorophosphate
HMPA	=	hexamethylphosphoramide
HOBt	=	1-hydroxybenzotriazole
HPLC	=	high performance liquid chromatography
HR-MAS	=	high resolution magic angle spinning
IBX	=	1-hydroxy-1,2-benziodoxol-3(1H)-one
ICP-AA	=	inductively coupled plasma atomic absorption
MAO	=	monoamine oxidase
MBHA	=	4-methylbenzhydrylamine
MeOTf	=	methyl trifluoromethanesulfonate
MOM	=	methoxymethyl
MW	=	microwave
NMO	=	4-methylmorpholine <i>N</i> -oxide
Oct	=	octanoate
pin	=	pinacol
PS	=	polystyrene
RCM	=	ring closing metathesis
REM	=	acryloyl Wang
Rink	=	4-(2',4'-dimethoxyphenyl-Fmoc-aminomethyl)-phenoxy
RT	=	room temperature
Sieber	=	9-Fmoc-amino-xanthen-3-yloxy
TBAF	=	tetrabutylammonium fluoride
TBDMS	=	<i>tert</i> -butyldimethylsilyl
TBSOTf	=	<i>tert</i> -butyldimethylsilyl trifluoromethanesulfonate

TFA	=	trifluoroacetic acid
TMEDA	=	<i>N,N,N',N'</i> -tetramethylethylenediamine
TMSCl	=	trimethylsilyl chloride
TMSOTf	=	trimethylsilyltriflate
TPAP	=	tetrapropylammonium perruthenate

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